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# Appendix A: Register Map

## .1 Conventions

Every custom MMIO block in this design is described in this appendix. The standard RISC-V CLINT (0x4000_1000), the DDR5 controller, the PCIe root complex, and the CXL host bridge follow their respective industry specifications and are not repeated here.

**Register access codes:** RO = read-only, RW = read-write, WO = write-only (read returns 0), W1C = write 1 to clear.

**Reset values** are the value after cold power-on before any firmware write, unless stated otherwise (some blocks have OTP-derived reset values that are programmed at factory test).

**Register width** is 32 bits unless stated otherwise. 64-bit registers occupy two consecutive 32-bit slots, low word at the lower offset.

**Reserved bits** must be written as 0 and are read as 0.

**Base addresses:** every block occupies a 4 KB page. Blocks with multiple instances (per-channel, per-bank, per-sensor) place instance $n$ at base $+n\times$ page size.

## .2 MMIO Base Address Map

| Base | Block | Instances | Products |
|---|---|---|---|
| 0x4000_0000 | GPIO | 1 | All |
| 0x4000_1000 | CLINT (standard) | 1 | All |
| 0x4000_2000 | PLIC-lite | 1 | All |
| 0x4000_3000 | Watchdog | 1 | All |
| 0x4000_4000 | Thermal | 4 (Ph-1), 6 (Pc-1), 8 (Dc-1S) | All |
| 0x4000_5000 | Host interface | 1 | All |
| 0x4000_6000 | DMA | 1 | All |
| 0x4000_7000 | Vector block | 1 | Ph-1, Pc-1 |
| 0x4000_8000 | OTP | 1 | All |
| 0x4000_9000 | Power status | 1 | All |
| 0x4000_A000 | Security engine | 1 | All |
| 0x4000_B000 | Debug authentication | 1 | All |
| 0x4000_C000 | CXL control | 1 | Pc-1, Dc-1S |
| 0x4000_D000 | NVL3 controller | 1 | All |

| Base | Block | Instances | Products |
|---|---|---|---|
| 0x4000_E000 | Crossbar controller | 1 | Ph-1, Pc-1 |

Writes to unimplemented addresses in a block's 4 KB page return 0 on read and are silently ignored on write (no bus error). This allows firmware to probe for block presence by writing a known pattern and reading back.

## .3 GPIO (0x4000_0000)

8 pins on Ph-1; 32 pins on Pc-1 and Dc-1S. Only the first 8 DATA/DIR/IE/IS/TRIG/IC/DRIVE/PULL/SLEW/I bits are meaningful on Ph-1.

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | DATA | RW | 0 | [n] pin n output value (only when DIR[n]=1) |
| 0x04 | DIR | RW | 0 | [n]=1 output, [n]=0 input |
| 0x08 | IE | RW | 0 | [n]=1 interrupt enabled |
| 0x0C | IS | RO | 0 | [n]=1 interrupt pending |
| 0x10 | TRIG | RW | 0 | [2n:2n+1]: 00=low, 01=high, 10=fall, 11=rise |
| 0x14 | IC | W1C | — | Write 1 to clear IS[n] |
| 0x18 | DRIVE | RW | 0x55555555 | [2n:2n+1]: 00=2 mA, 01=4 mA, 10=8 mA, 11=12 mA |
| 0x1C | PULL | RW | 0 | [2n:2n+1]: 00=none, 01=up, 10=down, 11=keeper |
| 0x20 | SLEW | RW | 0 | [n]=1 fast slew, [n]=0 slow |
| 0x24 | INSYNC | RW | 0 | [2n:2n+1]: 00=0, 01=2, 10=4, 11=8 cycle synchronizer delay |

On Ph-1, bits [31:8] of every register are reserved and read 0.

## .4 PLIC-lite (0x4000_2000)

Custom variant of the RISC-V PLIC. 16 interrupt sources on Ph-1; 64 on Pc-1 and Dc-1S. Sources are assigned in Appendix C. Register layout follows the standard RISC-V PLIC specification except where noted.

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x0000 | PRIORITY[0] | RW | 0 | Priority of source 0 (0=disabled) |
| ... | PRIORITY[n] | RW | 0 | 4 B per source; n = 0..N_SOURCE-1 |
| 0x1000 | PENDING | RO | 0 | 32-bit word per 32 sources; bit n = source n |

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x2000 | ENABLE | RW | 0 | 32-bit word per 32 sources; bit n = source n enabled for hart 0 |
| 0x200000 | THRESHOLD | RW | 0 | Hart 0 threshold (only hart 0 in this implementation) |
| 0x200004 | CLAIM | RW | — | Read returns highest-priority pending source; write same value to complete |
| 0x200008 | CFG | RW | 0 | [0]=enable PLIC; [1]=interrupt mode (0=level, 1=edge) |

Only hart 0 is supported; firmware running on other harts must route interrupts through hart 0 software interrupt (source 1). This is the single-hart-PLIC design trade-off.

## .5 Watchdog (0x4000_3000)

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | CTRL | RW | 0 | [0] ENABLE; [1] FEED (WO, self-clear); [2] PRESCALE_EN; [4:3] TIMEOUT_SEL (00=1 ms, 01=10 ms, 10=100 ms, 11=1 s); [5] IRQ_EN; [6] RESET_EN; [7] FORCE_RESET (WO); [8] CLR_RESET_CAUSE (WO) |
| 0x04 | STATUS | RO | 0x2 | [0] RUNNING; [1] READY; [2] TIMEOUT; [3] RESET_OCCURRED; [4] RS:SW; [5] RS:WDT; [6] RS:BOD; [7] RS:THERMAL; [8] RS:EXTERNAL |
| 0x08 | LOAD | RW | 0xFFFFFFFF | Timeout load (prescaler cycles) |
| 0x0C | COUNT | RO | 0xFFFFFFFF | Current count |

Reset cause bits [8:4] are latched from the last reset event and cleared only by writing CLR_RESET_CAUSE. They are the source of truth for boot-time diagnostic logging.

## .6 Thermal Sensors (0x4000_4000)

Per-sensor registers: instance $n$ at base $+n \times 0x100$. Number of sensors: 4 (Ph-1), 6 (Pc-1), 8 (Dc-1S).

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | RAW | RO | — | 16-bit signed raw count; convert with CAL_OFF + CAL_SLOPE |
| 0x04 | TEMP_mC | RO | — | Temperature in milli-°C, already calibrated (formula below) |
| 0x08 | CAL_OFF | RW | OTP | Calibration offset (OTP-programmed) |

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x0C | CAL_SLOPE | RW | OTP | Calibration slope (OTP-programmed) |
| 0x10 | THRESH | RW | 0x55F0 | Threshold in °C×100 (default 85°C×100 = 0x21F4) |
| 0x14 | HYST | RW | 500 | Hysteresis in °C×100 (default 5°C) |
| 0x18 | CTRL | RW | 0x1 | [0] ENABLE; [1] IRQ_EN; [2] TRIP_EN |
| 0x1C | STATUS | RO | 0 | [0] VALID; [1] ABOVE_THRESH; [2] TRIP_LATCHED (W1C) |
| 0x20 | TRIP_THRESH | RW | 0x5DC0 | Emergency shutdown threshold in °C×100 (default 115°C×100 = 0x5DC0) |

Conversion: ‘TEMP$_m C = (RAWCAL_SLOPE)/1024+CAL_OFF$‘.$The sensor is a bandgap PTAT type with 1C$ $-125 C after the two-point calibration.$

## .7 Host Interface (0x4000_5000)

SPI slave on Ph-1; PCIe endpoint config space on Pc-1 and Dc-1S. Registers below are common to both; per-product differences are noted.

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | CTRL | RW | 0 | [0] IF_ENABLE; [1] INT_MODE (SPI: level/edge; PCIe: MSI/MSI-X); [2] DMA_EN |
| 0x04 | STATUS | RO | 0 | [0] LINK_UP; [1] BUSY; [2] ERROR; [3] DMA_ACTIVE |
| 0x08 | IRQ_MASK | RW | 0 | Per-bit mask (mirrors PLIC source enable) |
| 0x0C | IRQ_PENDING | RO | 0 | Raw pending bits from PLIC (read-only mirror) |
| 0x10 | DMA_CMD_ADDR | RW | 0 | DMA descriptor list base address (host memory) |
| 0x14 | DMA_CMD_LEN | RW | 0 | Number of 16 B descriptors |
| 0x18 | DMA_CMD_TRIGGER | WO | — | Write 1 to start DMA descriptor processing |
| 0x1C | DMA_STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] ERROR; [3] ABORT |
| 0x20 | FW_VERSION | RO | OTP | Firmware version (OTP-programmed at factory) |
| 0x24 | FW_TRIGGER | WO | — | Write magic 0x4E564C33 to reload firmware image from NVL3 |
| 0x28 | PCIE_DEVICE_ID | RO | 0x106D | PCI device ID (Pc-1, Dc-1S only) |
| 0x2C | PCIE_VENDOR_ID | RO | 0x1D1F | PCI vendor ID (Pc-1, Dc-1S only) |

On Ph-1, offsets 0x28 and 0x2C read 0 (no PCIe). On Pc-1 and Dc-1S, offsets 0x00–0x24 are used for the internal DMA path; the standard PCIe config space at the PHY level (BARs, MSI-X table) is beyond this block’s scope.

## .8 DMA Controller (0x4000_6000)

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | CTRL | RW | 0 | [0] ENABLE; [1] LOOP_EN; [2] IRQ_EN |
| 0x04 | STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] ERROR; [3] ABORTED |
| 0x08 | SRC_ADDR | RW | 0 | Source address (low 32 bits) |
| 0x0C | SRC_ADDR_HI | RW | 0 | Source address (high 32 bits) |
| 0x10 | DST_ADDR | RW | 0 | Destination address (low 32 bits) |
| 0x14 | DST_ADDR_HI | RW | 0 | Destination address (high 32 bits) |
| 0x18 | LEN | RW | 0 | Transfer length in bytes (multiple of 8) |
| 0x1C | CHUNK | RW | 4096 | Chunk size in bytes (per burst) |
| 0x20 | TRIGGER | WO | — | Write 1 to start transfer |
| 0x24 | ABORT | WO | — | Write 1 to abort (DMA returns to idle within 1 µs) |
| 0x28 | CUR_ADDR | RO | 0 | Current source address (progress) |
| 0x2C | CUR_REMAIN | RO | 0 | Remaining bytes |
| 0x30 | ERROR_CODE | RO | 0 | [3:0] last error (1=src bus, 2=dst bus, 3=alignment, 4=timeout, 5=abort) |

One DMA engine, programmed from hart 0. Multiple channels are software-arbitrated via the LOOP_EN bit and a software descriptor queue in SRAM.

## .9 Vector Block Control (0x4000_7000)

Ph-1 and Pc-1 only. This block controls the analog crossbar's functional DAC/ADC chain (Part 5 §5.13 custom instructions trigger the compute; this block configures the chain).

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | VECMODE | RW | 0 | [1:0] mode: 00=idle, 01=2-bit, 10=4-bit (AO1), 11=8-bit (AO2) |
| 0x04 | DAC_CODE | RW | 0 | [3:0] DAC code for current row (used by crossbar.macc) |
| 0x08 | FIFO_WR | RW | — | Write 8 B input vector word (pushed to input FIFO) |
| 0x0C | FIFO_WR_COUNT | RO | 0 | Number of words currently in input FIFO |
| 0x10 | FIFO_RD | RO | — | Read 8 B output vector word (popped from output FIFO) |
| 0x14 | FIFO_RD_COUNT | RO | 0 | Number of words currently in output FIFO |
| 0x18 | FIFO_STATUS | RO | 0 | [0] IN_FULL; [1] IN_EMPTY; [2] OUT_FULL; [3] OUT_EMPTY; [4] OVERFLOW |
| 0x1C | CTRL | RW | 0 | [0] ENABLE; [1] AUTO_TRIGGER; [2] IRQ_EN |
| 0x20 | STATUS | RO | 0 | [0] BUSY; [1] MAC_DONE; [2] ERROR |

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x24 | TIA_GAIN | RW | 0 | [1:0] gain: 00=high (2.6 k), 01=mid, 10=low (10 k), 11=reserved |
| 0x28 | ADC_CONFIG | RW | 0 | [1:0] resolution: 00=8-bit, 01=10-bit (default) |
| 0x2C | CALIB_CTRL | RW | 0 | [0] START_CALIB; [1] ABORT_CALIB; [2] LOAD_FROM_OTP |
| 0x30 | CALIB_STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] ERROR; [3] OTP_LOADED |
| 0x34 | CALIB_COL | RW | 0 | Column index for calibration (0–511 or 0–1023) |
| 0x38 | CALIB_ROW | RW | 0 | Row index for calibration (0–511 or 0–1023) |

The four custom instructions (‘crossbar.ld‘, ‘.st‘, ‘.macc‘, ‘.wait‘) have hardware backing in this block: ‘.macc‘ triggers a MAC cycle, ‘.wait‘ polls MAC_DONE. Direct register access to FIFO_WR and FIFO_RD is used only during initialization and calibration.

## .10 OTP Controller (0x4000_8000)

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | ADDR | RW | 0 | OTP word address (32-bit words) |
| 0x04 | DATA_READ | RO | 0 | Data at ADDR (valid after READ_STROBE) |
| 0x08 | DATA_WRITE | RW | 0 | Data to write at ADDR |
| 0x0C | READ_STROBE | WO | — | Write 1 to start read; STATUS.RDY set when data valid |
| 0x10 | PROG_STROBE | WO | — | Write 1 to start program; STATUS.PROG_DONE set when complete |
| 0x14 | CTRL | RW | 0 | [0] PROG_ENABLE (must be 1 for program; requires VPP if fuse-based) |
| 0x18 | STATUS | RO | 0 | [0] RDY; [1] PROG_DONE; [2] PROG_FAILED; [3] LOCKED (OTP region is locked) |
| 0x1C | LOCK | WO | — | Write 1 to permanently lock OTP (irreversible) |
| 0x20 | LOCK_STATUS | RO | OTP | [n]=1 means region n locked; regions defined in §14.7 |
| 0x24 | READ_BACK | RO | 0 | Last-read value (for verification) |
| 0x28 | REGION_BASE | RW | 0 | [7:0] current region base (0=root key, 1=device, ..., 7=calibration) |

Programming sequence: write ADDR, write DATA_WRITE, write CTRL.PROG_ENABLE=1, write PROG_STROBE=1, poll STATUS.PROG_DONE (or PROG_FAILED), verify via READ_STROBE and compare to DATA_WRITE. Programming requires the OTP controller to have been unlocked at boot by the security engine (see Appendix B §B.7).

## .11 Power Status and Sequencing (0x4000_9000)

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | RAIL_STATUS | RO | 0 | [n]=1 rail n up. n=0 core, 1 I/O, 2 analog, 3 PLL, 4 LPDDR5X/DDR5, 5 auxiliary |
| 0x04 | RAIL_OK | RO | 0 | [n]=1 rail n within spec |
| 0x08 | SEQUENCE_STATE | RO | 0 | [3:0] state (0=off, 1=core, 2=io, 3=analog, 4=pll, 5=mem, 6=aux, 7=up, 8=shutdown) |
| 0x0C | BOD_STATUS | RO | 0 | [n]=1 brown-out detected on rail n (latched) |
| 0x10 | BOD_CLR | W1C | — | Clear BOD_STATUS |
| 0x14 | SHUTDOWN_TRIGGER | WO | — | Write 0xDEAD to trigger orderly shutdown |
| 0x18 | RESET_CAUSE | RO | 0 | Same encoding as Watchdog STATUS[8:4] |
| 0x1C | RESET_CLR | WO | — | Write 1 to clear RESET_CAUSE |
| 0x20 | DVFS_CTRL | RW | 0 | [3:0] target P-state (0=P0, 1=P1, ..., 4=P4); [8] REQUEST; [9] AUTO (hardware DVFS) |
| 0x24 | DVFS_STATUS | RO | 0 | [3:0] current P-state; [8] TRANSITIONING; [9] AUTO_ACTIVE |
| 0x28 | DVFS_TRIM_V | RW | 0 | [7:0] voltage trim (signed, 25 mV/LSB) |
| 0x2C | DVFS_TRIM_F | RW | 0 | [7:0] frequency trim (signed, 100 MHz/LSB) |
| 0x30 | THROTTLE | RO | 0 | [1:0] throttle level (0=none, 1=75 |

DVFS sequence: write DVFS_CTRL with target P-state and REQUEST=1; poll DVFS_STATUS until TRANSITIONING clears. Transition takes 5 µs nominal, 50 µs worst case (PLL relock + voltage ramp). The AUTO bit lets hardware choose the P-state from thermal and utilization; software DISABLEs AUTO by writing 0.

## .12 Security Engine (0x4000_A000)

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | AES_CTRL | RW | 0 | [0] START; [1] DECRYPT; [2] ENCRYPT; [3] KEY_SLOT (0–3); [4] MODE (0=XTS) |
| 0x04 | AES_STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] ERROR |
| 0x08 | AES_SRC_ADDR | RW | 0 | Source address (low) |
| 0x0C | AES_SRC_ADDR_H | RW | 0 | Source address (high) |
| 0x10 | AES_DST_ADDR | RW | 0 | Destination address (low) |
| 0x14 | AES_DST_ADDR_H | RW | 0 | Destination address (high) |
| 0x18 | AES_LEN | RW | 0 | Length in 16 B blocks |
| 0x1C | AES_TWEAK | RW | 0 | XTS tweak (16 B, at offset 0x1C–0x28) |
| ... | AES_TKEY | WO | — | Key slot register (write only, zero on read); 16 B per slot at 0x30 + slot×0x10 |

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x80 | SHA_CTRL | RW | 0 | [0] START; [1] RESET; [2] ALG (0=SHA-256, 1=SHA-384) |
| 0x84 | SHA_STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] ERROR |
| 0x88 | SHA_DATA | WO | — | 8 B data word |
| 0x8C | SHA_DIGEST_0 | RO | 0 | Result word 0 (of 8 or 12) |
| ... | SHA_DIGEST_n | RO | 0 | Result word n at 0x8C + n×4 |
| 0xA0 | ECDSA_CTRL | RW | 0 | [0] START_VERIFY; [1] START_SIGN |
| 0xA4 | ECDSA_STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] VERIFY_OK; [3] VERIFY_FAIL |
| 0xA8 | ECDSA_MSG_ADDR | RW | 0 | Message address (low) |
| 0xAC | ECDSA_MSG_ADDR_HI | RW | 0 | Message address (high) |
| 0xB0 | ECDSA_MSG_LEN | RW | 0 | Message length in bytes |
| 0xB4 | ECDSA_R | RO | 0 | Signature component r (32 B, at 0xB4–0xD3) |
| ... | ECDSA_S | RO | 0 | Signature component s (32 B) |
| 0xD8 | TRNG_CTRL | RW | 0 | [0] ENABLE |
| 0xDC | TRNG_DATA | RO | 0 | 4 random bytes per read (FIFO depth 64) |
| 0xE0 | TRNG_STATUS | RO | 0 | [0] READY; [1] HEALTH_FAIL |
| 0xE4 | DICE_CDI | RO | 0 | Compound Device Identifier (16 B at 0xE4–0xF3) |
| 0xF4 | DICE_MEASURE | RW | 0 | Next measurement source address (low) |
| 0xF8 | DICE_MEASURE_HI | RW | 0 | High 32 bits |
| 0xFC | DICE_CTRL | RW | 0 | [0] START; [1] UPDATE |
| 0x100 | DICE_STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] ERROR |

AES-XTS key slots are write-only and read back 0. Key material is loaded only at boot, from the ROM to the security engine via the OTP controller, never via software-visible paths. The DICE CDI is regenerated on every cold boot from the OTP device key + boot nonce + first measurement.

## .13 Debug Authentication (0x4000_B000)

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | CHALLENGE | RW | 0 | 32 B challenge written by debugger (write only; 0x00–0x1F) |
| 0x20 | RESPONSE | RO | 0 | 32 B response read by debugger (0x20–0x3F) |
| 0x40 | CTRL | RW | 0 | [0] GEN_RESPONSE; [1] VERIFY; [2] UNLOCK; [3] LOCK |
| 0x44 | STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] VERIFY_OK; [3] UNLOCKED; [4] TIMEOUT; [5] LOCKED |
| 0x48 | TIMEOUT | RW | 3600 | Unlock duration in seconds (default 3600 = 1 hour); 0 = unlimited |
| 0x4C | ATTEMPTS | RO | 0 | Failed unlock attempts (counter) |

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x50 | LOCK_DOWN | RO | 0 | [0]=1 if locked permanently after threshold attempts |

Unlock sequence: debugger generates random 32 B challenge, writes to CHALLENGE; hardware computes ECDSA signature of the challenge with the device-unique private key in OTP; debugger reads RESPONSE and verifies against the public key. If verification succeeds, debugger writes VERIFY=1 followed by UNLOCK=1; hardware opens JTAG for TIMEOUT seconds. Failed verifications increment ATTEMPTS; after 5 failed attempts, LOCK_DOWN is set and unlock requires factory RMA.

## .14 CXL Control (0x4000_C000)

Pc-1 and Dc-1S only. Base registers; the full standard CXL config space is separate.

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | LINK_STATUS | RO | 0 | [0] UP; [1] TRAINED; [2] RETRAIN_REQ; [3] ERROR; [7:4] speed |
| 0x04 | LINK_CTRL | RW | 0 | [0] ENABLE; [1] FORCE_RETRAIN; [2] RESET_LINK |
| 0x08 | LINK_ERROR | RO | 0 | [0] CRC_ERR; [1] REPLAY_ERR; [2] MLF_ERR; [3] TRAIN_ERR (W1C) |
| 0x0C | LINK_ERROR_CLR | W1C | — | Clear bits in LINK_ERROR |
| 0x10 | LANE_STATUS | RO | 0 | [31:0] per-lane status (bit n = lane n trained) |
| 0x14 | SPEED | RW | 0x40 | [7:0] target speed (0x40 = 64 GT/s, 0x20 = 32 GT/s, 0x10 = 16 GT/s) |
| 0x18 | WIDTH | RW | 0x10 | [7:0] target width (0x10 = x16) |
| 0x1C | NUMA_NODE_ID | RW | 0x0 | This socket's NUMA node ID (used by ACPI SRAT) |
| 0x20 | REMOTE_SOCKET_COUNT | RO | 0 | Number of remote sockets discovered at boot |
| 0x24 | REMOTE_SOCKET_SEL | RW | 0 | Index into remote socket table (0..N-1) |
| 0x28 | REMOTE_NUMA_NODE | RW | 0 | NUMA node ID of remote socket (per index) |
| 0x2C | INTERLEAVE | RW | 0 | Memory interleave mode (0=none, 1=2-way, 2=4-way) |
| 0x30 | INTERLEAVE_GRAN | RW | 0x100 | Interleave granularity in bytes (default 256) |

The CXL block controls the link and NUMA topology only. The actual memory window (for CXL.mem targets) is configured via the standard CXL config space at 0x0000_0003_xxxx_xxxx; that path is documented in the CXL 3.0 specification and not repeated here.

## .15 NVL3 Controller (0x4000_D000)

The NVL3 controller manages the FeFET non-volatile L3 array. The data window at 0x0000_0000_8000_0000 routes cache-line reads and writes through the controller's row/page logic; the control registers below configure the controller.

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | CTRL | RW | 0 | [0] ENABLE; [1] START_WRITE_VERIFY; [2] START_ERASE; [3] START_PRE_WAKEUP; [4] RESET; [5] WRITE_BUF_EN |
| 0x04 | STATUS | RO | 0 | [0] READY; [1] BUSY; [2] ERROR; [3] VERIFY_FAILED; [4] ENDURANCE_WARN; [5] CALIB_IN_PROGRESS |
| 0x08 | ROW_ADDR | RW | 0 | Row address (0–4095) |
| 0x0C | BANK_ADDR | RW | 0 | Bank address (0–255) |
| 0x10 | DATA_LOW | RW | 0 | 256 bits of write data, low (offsets 0x10–0x2F) |
| 0x30 | DATA_HIGH | RW | 0 | 256 bits of write data, high (offsets 0x30–0x4F) |
| 0x50 | RDATA_LOW | RO | 0 | 256 bits of read data, low |
| 0x70 | RDATA_HIGH | RO | 0 | 256 bits of read data, high |
| 0x90 | WRITE_VERIFY_CTRL | RW | 0x302 | [3:0] max retries (default 3); [11:8] pulse width (default 2 = 25 ns) |
| 0x94 | WRITE_VERIFY_STATUS | RO | 0 | [7:0] attempt count; [8] SUCCESS; [9] FAILED |
| 0x98 | ECC_STATUS | RO | 0 | [0] SINGLE_BIT; [1] DOUBLE_BIT; [8] OVERFLOW |
| 0x9C | ECC_ADDR | RO | 0 | Address of the last ECC event (row ¡¡ 16 — bank) |
| 0xA0 | BAD_BLOCK | RO | 0 | [15:0] bad-block count; [16] last-bad-block-address-available |
| 0xA4 | BAD_BLOCK_ADDR | RO | 0 | Last bad-block address |
| 0xA8 | WEAR_COUNT_LOW | RO | 0 | Cumulative write count (low 32) |
| 0xAC | WEAR_COUNT_HIGH | RO | 0 | Cumulative write count (high 32) |
| 0xB0 | PRE_WAKEUP_CTRL | RW | 0x3E8 | [15:0] pulse count per cell (default 1000 = 0x3E8) |
| 0xB4 | PRE_WAKEUP_STATUS | RO | 0 | [0] DONE; [1] BUSY; [2] ERROR |
| 0xB8 | CAL_TABLE_BASE | RW | OTP | Calibration table base address in SRAM |
| 0xBC | CAL_TABLE_LEN | RW | OTP | Calibration table length in 4 B words |
| 0xC0 | REF_OFFSET_BASE | RW | OTP | Per-column reference offset base (63–167 KB table) |
| 0xC4 | CALIB_START | WO | — | Write 1 to (re)load calibration from OTP |
| 0xC8 | CALIB_STATUS | RO | 0 | [0] DONE; [1] BUSY; [2] ERROR |
| 0xCC | ERROR_LOG | RO | 0 | [7:0] last error code (1=verify, 2=ecc, 3=endurance, 4=bad-block-limit, 5=calibration) |

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0xD0 | IRQ_ENABLE | RW | 0 | [0] ECC_SB; [1] ECC_DB; [2] VERIFY; [3] ENDURANCE; [4] BAD_BLOCK |
| 0xD4 | IRQ_STATUS | RO | 0 | Mirrors IRQ_ENABLE bit layout; latched until IC write |
| 0xD8 | IC | W1C | — | Clear bits in IRQ_STATUS |

Write sequence (single row): write ROW_ADDR, BANK_ADDR, DATA_LOW (8×32 bits), DATA_HIGH (8×32 bits); write CTRL.START_WRITE_VERIFY; poll STATUS.BUSY or wait for VERIFY interrupt; check WRITE_VERIFY_STATUS.SUCCESS (or FAILED). The controller internally performs erase, program, read-verify, and retry up to max-retries.

Read sequence (single row): write ROW_ADDR, BANK_ADDR; write CTRL.START_WRITE_VERIFY is NOT used for reads; instead, reads are triggered by the data-window access at 0x8000_0000. Alternatively, direct controller read: write ROW_ADDR, BANK_ADDR, then read RDATA_LOW / RDATA_HIGH after the controller's page-open latency. The controller's direct read path is provided for bring-up and ECC scrub.

Pre-wake-up: write PRE_WAKEUP_CTRL with the desired pulse count (default 1000); write CTRL.START_PRE_WAKEUP; poll PRE_WAKEUP_STATUS.DONE. Pre-wake-up must be applied to every bank before the NVL3 data region is used as persistent storage. Factory test applies $10^3 pulses to the NVL3 data region and 10^4 to the microcode region$.

Calibration: at factory test and after any NVL3 configuration change, write CALIB_START; poll CALIB_STATUS.DONE. The controller reads the calibration table from OTP, populates REF_OFFSET, and validates each column's reference against a known pattern.

## .16 Crossbar Controller (0x4000_E000)

Ph-1 and Pc-1 only. Full functional configuration of the analog crossbar. The compute path is the custom-instruction interface (‘crossbar.macc‘ etc.); this block configures the analog chain and manages weight updates and calibration.

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x00 | CTRL | RW | 0 | [0] ENABLE; [1] W+/W-MODE (0=Option B time-mux, 1=Option A parallel); [2] AUTO_DAC; [3] IRQ_EN |
| 0x04 | STATUS | RO | 0 | [0] READY; [1] BUSY; [2] ERROR; [3] MAC_DONE; [4] CALIB_BUSY |
| 0x08 | W_UPDATE_CTRL | RW | 0 | [0] START; [1] ABORT; [2] VERIFY_EN; [3] HALT_ON_ERR |
| 0x0C | W_UPDATE_STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] ERROR; [3] VERIFIED |
| 0x10 | W_SRC_ADDR | RW | 0 | Weight source address in SRAM (low) |
| 0x14 | W_SRC_ADDR_HI | RW | 0 | High 32 bits |
| 0x18 | W_LEN | RW | 0 | Length in cells |
| 0x1C | W_ROW | RW | 0 | Row to update (0–511 or 0–1023) |
| 0x20 | W_COL | RW | 0 | Column start |
| 0x24 | W_PATTERN | RW | 0 | Current weight pattern (valid after update) |
| 0x28 | W_VERIFY | RO | 0 | Read-back value of last updated cell |

| Offset | Name | Access | Reset | Bit fields |
|---|---|---|---|---|
| 0x2C | CALIB_CTRL | RW | 0 | [0] START; [1] ABORT; [2] LOAD_FROM_OTP; [3] TEMPCO_EN |
| 0x30 | CALIB_STATUS | RO | 0 | [0] BUSY; [1] DONE; [2] ERROR; [3] OTP_LOADED |
| 0x34 | CALIB_COL | RW | 0 | Column to calibrate (0–511 or 0–1023) |
| 0x38 | CALIB_GAIN_COARSE | RW | 0 | Coarse gain trim (0–7) |
| 0x3C | CALIB_GAIN_FINE | RW | 0 | Fine gain trim (0–255) |
| 0x40 | CALIB_OFFSET | RW | 0 | Signed offset trim (-2048..+2047) |
| 0x44 | TEMPCO_BASE | RW | OTP | Temperature coefficient table base |
| 0x48 | TEMPCO_LEN | RW | OTP | Length in 4 B words |
| 0x4C | ADC_TRIM | RW | OTP | ADC trim (INL, DNL) from OTP |
| 0x50 | DAC_TRIM | RW | OTP | DAC trim (offset, gain) from OTP |
| 0x54 | TIA_TRIM | RW | OTP | TIA trim (offset, bandwidth) from OTP |
| 0x58 | IRQ_ENABLE | RW | 0 | [0] MAC_DONE; [1] W_UPDATE; [2] CALIB |
| 0x5C | IRQ_STATUS | RO | 0 | Mirrors IRQ_ENABLE bit layout; latched until IC write |
| 0x60 | IC | W1C | — | Clear bits in IRQ_STATUS |

Weight update sequence: write W_SRC_ADDR (weight pattern in SRAM), W_LEN, W_ROW, W_COL; write W_UPDATE_CTRL.START=1; poll W_UPDATE_STATUS.BUSY or wait for W_UPDATE interrupt; check W_UPDATE_STATUS.DONE and .VERIFIED. The controller programs FeFET_A and FeFET_B in the row, applies read-verify per cell, retries up to 3 times.

Calibration: write CALIB_COL and optionally CALIB_GAIN_COARSE / CALIB_GAIN_FINE / CALIB_OFFSET overrides; write CALIB_CTRL.START=1; poll CALIB_STATUS.DONE. Factory calibration cycles all columns and stores the resulting per-column gain/offset in OTP. At boot, write CALIB_CTRL.LOAD_FROM_OTP=1 to restore.

Option A/B mode: at boot, firmware reads the product's config from OTP (or from an ID register not shown here) and writes CTRL.W+/W-MODE appropriately. On Ph-1 (single crossbar), mode=0 (time-multiplexed); on Pc-1 (parallel pair), mode=1.

# Appendix B: Firmware Interface Specification

This appendix specifies the protocols between firmware and hardware for the sequences that the register map alone does not fully define: NVL3 row write-verify, NVL3 pre-wake-up, crossbar weight update, crossbar calibration, firmware update, security engine key provisioning, and error recovery.

Every sequence is stated as a step-numbered procedure. Every register mentioned is defined in Appendix A. Timing budgets are stated where the sequence has a hard real-time constraint.

## .17 Conventions

**Status polling:** where the text says "poll STATUS.X", firmware reads STATUS in a loop until the named bit is set or a timeout (default 100 ms) expires. Timeout handling: escalate to the error recovery sequence at §B.8.

**Interrupt alternative:** wherever a status-poll is described, firmware may enable the corresponding IRQ_ENABLE bit and service the interrupt. The protocols below state the polling form because it is the simplest; interrupt-driven versions follow the same register writes but replace the poll with an interrupt handler.

**Ordering:** all register writes to a single block must be separated by at least one ‘fence‘ instruction to ensure MMIO ordering. Writes to different blocks have no ordering constraint unless stated.

**Memory barriers:** where a register write triggers a DMA transfer, the source buffer must be flushed to memory (or marked non-cacheable) before the trigger. This is the standard RISC-V ‘fence rw, rw‘ before writing the trigger register.

## .18 NVL3 Row Write-Verify

Purpose: program a single 512-bit row of the NVL3 with a 512-bit data pattern, with automatic read-verify and retry.

**Preconditions:** NVL3 controller enabled (CTRL.ENABLE=1); the row is not currently in use as NVL3 cache; the caller has exclusive access to the bank (typically the NVL3 driver, during a cache line eviction or a boot-time firmware write).

**Procedure:**

1. Write ROW_ADDR with the target row (0–4095).
2. Write BANK_ADDR with the target bank (0–255 for Ph-1/Pc-1, 0–95 for Dc-1S).
3. Write DATA_LOW (8×32-bit words, 0x10–0x2F) with the low 256 bits of the row.
4. Write DATA_HIGH (8×32-bit words, 0x30–0x4F) with the high 256 bits of the row.
5. Write WRITE_VERIFY_CTRL to set max-retries (default 3) and pulse width (default 25 ns).
6. Write CTRL.START_WRITE_VERIFY=1. This triggers the row erase + program + read-verify sequence.
7. Poll STATUS.BUSY until it clears (typical 400 ns, max 1.2 µs).
8. Read WRITE_VERIFY_STATUS.SUCCESS. If 1, proceed to step 9. If 0, read WRITE_VERIFY_STATUS.a and STATUS.ERROR; then either retry (re-write CTRL.START_WRITE_VERIFY=1) or invoke error recovery (§B.8).
9. Read ECC_STATUS. If SINGLE_BIT=1, log the address (ECC_ADDR) and clear via IC.SINGLE_BIT=1. If DOUBLE_BIT=1, mark the row bad (BAD_BLOCK) and return error.
10. Return success.

**Timing:** 400 ns nominal for a single-iteration write; up to 1.2 µs worst-case (4 iterations). The controller performs the verify read back from the FeFET cell, not from cache.

**Interrupt path:** enable IRQ_ENABLE.VERIFY=1; the handler reads WRITE_VERIFY_STATUS and STATUS and follows the same disposition logic.

**Cache interaction:** this sequence writes directly to NVL3. If the target row is currently cached in SRAM L3 or L2, the caller must invalidate those lines before starting (or accept that the next access will be a miss and re-fill from NVL3).

## .19 NVL3 Pre-Wake-Up

Purpose: apply the HZO polarization wake-up pulse sequence to a bank or to the full NVL3, so that the persistent retention target is met.

**Preconditions:** the target region is currently erased (all zeros) or can be treated as uninitialized. Pre-wake-up writes a dummy pattern, so any prior data is lost.

**Procedure (per bank):**

1. Write BANK_ADDR.
2. Write PRE_WAKEUP_CTRL with the pulse count. Factory default: 1000 pulses for the NVL3 data region, 10000 for the microcode region. Write 0x3E8 for data, 0x2710 for microcode.
3. Write CTRL.START_PRE_WAKEUP=1.
4. Poll PRE_WAKEUP_STATUS.DONE. The controller applies the pulse train in a single sweep; typical time 1.3 s for 1000 pulses (25 ns per pulse plus setup overhead), 13 s for 10000 pulses.
5. Read PRE_WAKEUP_STATUS.ERROR. If 1, error recovery; if 0, return success.

**Timing:** 1.3 s per bank for 1000 pulses. The full NVL3 (256 banks for Ph-1/Pc-1, 96 for Dc-1S) takes 333 s (Ph-1) or 125 s (Dc-1S) at factory test. This is why pre-wake-up is done once at factory and not at every boot.

**After pre-wake-up:** the affected region must be re-initialized (factory test writes the microcode image and the initial NVL3 content) and re-calibrated (write CALIB_START=1 and poll CALIB_STATUS.DONE).

## .20 NVL3 Calibration Load and Validation

Purpose: on each cold boot, load the per-column calibration table from OTP into the controller's working SRAM, and validate that a known pattern reads back correctly with the loaded offsets.

**Preconditions:** OTP controller has been unlocked by the security engine; OTP calibration region is populated (factory-programmed).

**Procedure:**

1. Write CALIB_START=1. The controller reads the calibration table from OTP (base and length are OTP-programmed into CAL_TABLE_BASE and CAL_TABLE_LEN at factory).
2. Poll CALIB_STATUS.DONE (typical 5 ms, max 20 ms).
3. Read CALIB_STATUS.ERROR. If 1, error recovery (§B.8).
4. Validation: write a known pattern to bank 0, row 0; read back; compare; if mismatch, error recovery.
5. If validation succeeds, return success. The NVL3 controller is now ready to serve the data window at 0x8000_0000.

**Boot path requirement:** this sequence must complete before any NVL3 access, including firmware image load. Boot ROM calls it as its first action after PLL lock.

## .21 Crossbar Weight Update

Purpose: program a contiguous run of weight cells in the analog crossbar to a new pattern (per-model weight update).

**Preconditions:** crossbar controller enabled (CTRL.ENABLE=1); the target cells are currently unused or are being re-purposed; the new pattern is in SRAM at a firmware-controlled address.

**Procedure:**

1. Write W_SRC_ADDR (low) and W_SRC_ADDR_HI (high) with the SRAM address of the new weight pattern. The pattern is a contiguous array of 2-bit (or 4-bit or 8-bit, in the AO configurations) weight values, packed according to the current VECMODE.
2. Write W_LEN with the number of cells to update. Maximum per call: 512 (one row of a $512^2$ crossbar) or 1024 (one row of a $1024^2$ crossbar).
3. Write W_ROW with the target row.
4. Write W_COL with the target column start.
5. Write W_UPDATE_CTRL.START=1, .VERIFY_EN=1, .HALT_ON_ERR=0 (or 1 if the caller can tolerate a partial update).
6. Poll W_UPDATE_STATUS.BUSY (typical 38 µs for a 512-cell row on Ph-1, 77 µs for a 1024-cell row on Pc-1; these are worst-case values from §5.12).
7. Read W_UPDATE_STATUS.DONE and .VERIFIED. If VERIFIED=1, the update succeeded for all cells. If VERIFIED=0, read W_VERIFY (the value of the last cell) and the source data at the corresponding offset; identify the first failing cell and either retry (re-write the failing segment) or invoke error recovery.
8. Return success.

**Ordering:** multiple weight updates must be serialized (wait for one to complete before starting another). Firmware enforces this via a mutex in the AI driver.

**Inference suspension:** during a weight update, inference must be suspended (no ‘crossbar.macc‘ issued). The AI driver enforces this via its own scheduling; the hardware does not enforce it, and issuing MAC during an update produces undefined results.

## .22 Crossbar Calibration

Purpose: measure and store the per-column gain, offset, and temperature coefficient of the crossbar, so that the separable model in §5.9 has valid coefficients.

**Procedure (per column):**

1. For each of the four known weight states (00, 01, 10, 11), program the entire column with that weight (W_UPDATE with W_ROW=0, W_COL=all, W_LEN=all).
2. For each of the 16 DAC codes, apply the DAC code as the input to the entire column, read the ADC output for each row, and compute the column-averaged current.
3. The result is a $4 \times 16$ table of (weight_state, dac_code) $\rightarrow$ column_average_current.
4. Fit the separable model: gain = slope of (current vs weight_state at fixed DAC); offset = intercept; tempco = the difference between the 25°C and 85°C fits divided by 60°C.

5. Store the fitted parameters in OTP via the OTP controller (Appendix A §A.10). Per column: 2 bits weight-state-major + 20 bits gain + 2 bits tempco per the schema in §5.9.

**Time:** approximately 20 minutes per die for a $1024^2$ crossbar at factory test; this is the dominant crossbar factory test time. The full procedure is described in the Crossbar Test Program (separate artifact).

**Boot path:** at cold boot, firmware writes CALIB_CTRL.LOAD_FROM_OTP=1 and polls CALIB_STATUS.OTP_LOADED; this restores the fitted coefficients into the working registers.

## .23 Firmware Update (Dual-Bank A/B)

Purpose: atomically replace the running firmware with a new signed image, with rollback on failure.

**Preconditions:** new image is signed by the OEM ECDSA P-256 key and has a version number greater than the current running version (rollback protection).

**Procedure:**

1. Firmware reads the new image into SRAM (from host, SPI, or storage) and verifies its ECDSA P-256 signature against the OEM public key in OTP.
2. If verification fails, abort and log.
3. If verification succeeds, firmware writes the new image to the backup NVL3 bank. The backup bank is the one that firmware is NOT currently running from (A/B alternation).
4. After the write completes and read-verifies, firmware writes the new image's version number to the OTP rollback counter area, using the OTP controller.
5. Firmware writes the boot pointer in OTP to point at the backup bank, then resets.
6. On the next boot, Boot ROM reads the boot pointer, verifies the selected bank's signature, and jumps to it.
7. If Boot ROM fails to verify or if the kernel fails to reach userspace within a watchdog timeout (typically 60 s), Boot ROM reverts the boot pointer to the previous bank and resets. The previous bank is retained for exactly this purpose.

**Physical presence:** unsigned recovery images (for RMA only) require the physical-presence button to be held during reset. The button state is read via a GPIO input; firmware enforces the policy.

**Anti-rollback:** firmware compares the new image's version number to the OTP rollback counter. A newer version always wins; a same-or-older version is rejected unless physical presence is asserted.

## .24 Security Engine Key Provisioning

Purpose: install the device-unique ECDSA key, the OEM root key hash, and the debug public key into OTP at factory test. This is the only time these values are written; they are then locked.

**Preconditions:** OTP is unlocked (LOCK_STATUS=0); the security engine has authenticated the factory programmer via a factory-level challenge/response (separate flow, not described here); the keys are provided as files by the OEM.

**Procedure:**

1. Write REGION_BASE=0 (root key hash region).
2. Write ADDR=0, DATA_WRITE=first 32 bits of root key hash, CTRL.PROG_ENABLE=1, PROG_STROBE=1.
3. Poll STATUS.PROG_DONE (typical 100 µs per word; a 32 B hash is 8 words, 800 µs total).
4. Repeat for the full root key hash (32 B = 8 words).
5. Repeat the read-verify for each word (READ_STROBE, compare DATA_READ to DATA_WRITE).
6. Write REGION_BASE=1 (device key region). Repeat with the 32 B device ECDSA private key.
7. Write REGION_BASE=2 (debug public key region). Repeat with the 32 B debug public key.
8. Write REGION_BASE=3 (rollback counter region). Initialize the counter to 1.
9. Write REGION_BASE=4 (boot nonce region). Initialize the nonce to a random 16 B value generated by the TRNG.
10. For each region, write LOCK=1 and poll LOCK_STATUS until the region shows locked.
11. The security engine now enforces that OTP writes to these regions fail (STATUS.LOCKED=1) and that the security boot flow uses the provisioned values.

**Failure handling:** if any read-verify fails (DATA_READ  DATA_WRITE), the device is scrapped. There is no retry on OTP programming. This is why the security provisioning step is done last, after all other factory tests pass.

**Backup key handling:** the OEM's factory test program holds the keys in memory during provisioning and zeroizes them on completion. The keys are not stored on the factory test system between runs.

## .25 Error Recovery

Purpose: given any error status from any block, restore the system to a defined safe state and, where possible, recover without a reset.

**NVL3 verify failed:** read WRITE_VERIFY_STATUS; if attempt count ¡ max-retries, retry (re-write CTRL.START_WRITE_VERIFY=1). If max-retries exhausted, mark the row bad: write BAD_BLOCK (increment count), write BAD_BLOCK_ADDR=current row; the NVL3 driver then excludes the row from use and remaps to a redundant row (software-level remapping; the hardware does not auto-repair). Return error to caller.

**NVL3 ECC double-bit:** read ECC_ADDR; read back the row from a redundant copy (if a redundant copy exists) or from the source of truth (SRAM or DDR5); log the address; mark the row bad as above.

**NVL3 endurance warning:** read WEAR_COUNT_LOW/HIGH; if the count exceeds the endurance threshold (default: 50

**Crossbar weight update failed:** read W_VERIFY and compare against the source; retry the failing segment. If 3 consecutive retries fail, mark the row bad (log for factory RMA) and continue with the rest of the model.

**Crossbar calibration failed:** read CALIB_STATUS; if ERROR=1, the calibration table in OTP is corrupt or the on-die reference is out of range. Retry once; if it still fails, boot without crossbar (set VECMODE=00 idle) and log the fault.

**Thermal trip:** read THROTTLE and the trip sensor's raw count; if thermal shutdown is asserted, no software recovery is possible; the PMIC shuts the rails down. Firmware must not attempt to re-enable. On next boot, read the trip sensor's STATUS.TRIP_LATCHED to see the cause.

**Watchdog reset:** on next boot, read RESET_CAUSE. If RS:WDT, the firmware that was running is suspect; boot into a safe recovery image (a separate signed image at a fixed OTP address) that logs the incident, then reboots to the primary image. If the safe image also fails, boot into the factory diagnostic image (physical presence required).

**PCIe/CXL link error:** read LINK_ERROR; write LINK_ERROR_CLR; if the error recurs within 100 ms, force a retrain (write LINK_CTRL.FORCE_RETRAIN=1) and wait for LINK_STATUS.UP. If retrain fails 3 times, fall back to reduced speed (write SPEED=0x20 or 0x10) and continue.

**Brown-out:** read BOD_STATUS to identify which rail dropped; report to BMC (Pc-1, Dc-1S) or log to NVL3 (Ph-1); do not attempt to continue until the rail is stable.

## .26 Summary Table: Firmware Sequences

| Sequence | When | Typical time | Failure action |
|---|---|---|---|
| NVL3 row write-verify | On cache eviction or explicit write | 400 ns | Retry up to 3, then mark row bad |
| NVL3 pre-wake-up | Factory once; on NVL3 re-init | 1.3 s/bank | Read STATUS; error recovery |
| NVL3 calibration | Every cold boot | 5 ms | Boot into safe image |
| Crossbar weight update | Per model change | 38–77 µs | Retry segment; mark row bad |
| Crossbar calibration | Factory once; on die re-init | 20 min | Boot without crossbar |
| Firmware update | On new image | Variable | Watchdog rollback |
| Security key provisioning | Factory once | ¡10 ms | Scrap device |
| Error recovery | On any error | Variable | See §B.8 |

# Appendix C: Interrupt Source Assignments

This appendix assigns the PLIC-lite interrupt sources. 16 sources are available on Ph-1; 64 on Pc-1 and Dc-1S. Sources 0–15 are common to all products; sources 16–63 exist only on Pc-1 and Dc-1S.

## .27 Source Assignment Table

| Source | Signal | Block | Edge/Level | Priority (default) |
|---|---|---|---|---|
| 0 | (reserved, tie 0) | — | — | — |
| 1 | Software interrupt (MSIP) | CLINT | level | 7 |
| 2 | Timer interrupt (MTIP) | CLINT | level | 7 |
| 3 | External (MEIP) | CLINT | level | 7 |
| 4 | NVL3 ECC single-bit | NVL3 | level | 5 |
| 5 | NVL3 ECC double-bit | NVL3 | level | 6 |
| 6 | NVL3 write-verify failed | NVL3 | level | 5 |
| 7 | NVL3 endurance warning | NVL3 | level | 4 |
| 8 | NVL3 bad-block limit | NVL3 | level | 4 |
| 9 | Watchdog timeout | Watchdog | level | 7 |
| 10 | Thermal high (85°C) | Thermal | level | 6 |
| 11 | Thermal trip (115°C) | Thermal | level | 7 |
| 12 | Brown-out | Power | level | 7 |
| 13 | DMA complete | DMA | level | 3 |
| 14 | Host interface (SPI/PCIe) | Host | level | 4 |
| 15 | Debug unlock timeout | Debug | level | 3 |
| 16 | Crossbar MAC done | Crossbar | level | 3 |
| 17 | Crossbar weight update done | Crossbar | level | 4 |
| 18 | Crossbar calibration done | Crossbar | level | 4 |
| 19 | Vector FIFO underflow/overflow | Vector | level | 3 |
| 20 | GPIO interrupt | GPIO | level | 3 |
| 21 | Security engine AES done | Security | level | 5 |
| 22 | Security engine SHA done | Security | level | 5 |
| 23 | Security engine ECDSA done | Security | level | 5 |
| 24 | TRNG health failure | Security | level | 6 |
| 25 | DICE measurement complete | Security | level | 5 |
| 26 | OTP read complete | OTP | level | 3 |
| 27 | OTP program failed | OTP | level | 6 |
| 28 | CXL link up | CXL | level | 5 |
| 29 | CXL link down | CXL | level | 6 |
| 30 | CXL link error | CXL | level | 5 |
| 31 | PCIe MSI (aggregated) | Host | level | 4 |
| 32 | DDR5 controller ECC event | DDR5 | level | 5 |
| 33 | DDR5 training complete | DDR5 | level | 4 |
| 34 | SRAM L3 ECC single-bit | L3 | level | 4 |

| Source | Signal | Block | Edge/Level | Priority (default) |
|---|---|---|---|---|
| 35 | SRAM L3 ECC double-bit | L3 | level | 6 |
| 36 | L2 ECC single-bit | L2 | level | 4 |
| 37 | L2 ECC double-bit | L2 | level | 6 |
| 38 | L1 ECC single-bit | L1 | level | 3 |
| 39 | L1 ECC double-bit | L1 | level | 5 |
| 40 | MMU TLB shootdown | MMU | level | 4 |
| 41 | Branch predictor parity | Core | level | 3 |
| 42 | ROB parity | Core | level | 4 |
| 43 | Register file parity | Core | level | 4 |
| 44 | Pipeline lockstep mismatch | Core | level | 7 |
| 45 | PMU overflow | PMU | level | 2 |
| 46 | RISC-V custom instruction error | Core | level | 3 |
| 47 | JTAG debug event | Debug | level | 2 |
| 48 | (reserved) | — | — | — |
| ... | (reserved 48–62) | — | — | — |
| 63 | NMI (aggregated) | PLIC | level | 7 |

## .28 Priority Encoding

Priority values are 0–7 (3-bit priority, matching the PLIC-lite 4-bit priority field with the top bit reserved). 0 = disabled (source cannot interrupt). 7 = highest. Firmware may override the default priorities at boot by writing PLIC PRIORITY[n]. The defaults above are chosen so that the highest-severity sources (brown-out, watchdog, thermal trip, NMI, pipeline lockstep mismatch) are 7; the "attention" sources (ECC double-bit, security health fail) are 6; the "log-and-continue" sources are 3–5; the "informational" sources (DMA complete, PMU overflow) are 2–3.

## .29 Per-Product Differences

**Ph-1:** sources 16–63 are not implemented. Crossbar and CXL sources do not exist. Sources 34–47 are also not present (SRAM/caches are the same, but no separate register file parity interrupt on the 12-wide phone core — the parity is checked but not latched to PLIC). The PLIC-lite on Ph-1 has 16 usable sources (0–15, with 0 reserved, so 15 usable).

**Pc-1:** all 64 sources implemented.

**Dc-1S:** all 64 sources implemented. Crossbar sources 16–18 read as always-zero (no crossbar on Dc-1S), but the lines exist and are valid to enable — they simply never fire.

## .30 Interrupt Handler Ordering

When multiple sources are pending, the PLIC-lite serves them in strict priority order (highest priority first; ties broken by lowest source number). Firmware may reorder by writing PRIORITY.

For latency-critical sources (brown-out, watchdog), firmware should also configure the watchdog's own RESET_EN bit as a hardware backstop: if firmware does not service the

watchdog interrupt within the TIMEOUT window, the watchdog resets the chip. This is the standard two-level defense (interrupt for logging, reset for safety).

### .31 NMI Configuration

Source 63 (NMI) is the aggregated non-maskable interrupt. It fires when any of the following occurs: thermal trip (11), pipeline lockstep mismatch (44), or brown-out (12). The NMI handler in firmware must immediately:

1. Read all trip-status registers (THERMAL.STATUS, BOD_STATUS, pipeline parity) to identify the cause.
2. Log to NVL3 (if available) or to BMC (if available).
3. Initiate the safe-state sequence: stop all DMA, quiesce the crossbar, halt user cores, and prepare for reset.
4. Trigger watchdog FORCE_RESET=1 if the safe-state sequence does not complete within 100 ms.

The NMI cannot be masked by PLIC settings. It is guaranteed to fire on the next cycle after the event.

# Appendix D: Device Tree and ACPI Bindings

This appendix specifies the platform description data that firmware presents to the operating system. The design supports two boot-time platform description modes: Device Tree (DT), standard for Linux and Android on RISC-V; and ACPI, standard for NUMA-aware server Linux, CXL-aware operating systems, and other ACPI-dependent software. The two are mutually exclusive at boot; the boot ROM selects one based on an OTP configuration bit and passes the corresponding pointer to the OS.

This appendix specifies:

- The boot modes and how firmware chooses between them.
- The Device Tree bindings for every custom block (compatible strings, node structure, required and optional properties).
- The ACPI table inventory and the per-product values that populate each table.
- The DSDT namespace for every custom block (full ASL definitions).
- The boot handoff contract (registers, memory state, state transitions).
- The validation procedure that verifies the tables are correct before tape-out.

The standard ACPI table formats (RSDP, XSDT, MADT, SRAT, SLIT, MCFG, SPCR, GTDT, RHCT, IORT, CEDT, PPTT) are defined in ACPI 6.5 and are not repeated here; this appendix gives the design-specific values and the DSDT namespace. The standard DT bindings for RISC-V (PLIC, CLINT, CPU, memory) are defined in the Linux kernel documentation; this appendix gives the design-specific nodes and properties.

## .32 Boot Modes

Two boot modes are supported. The mode is selected by an OTP bit (OTP region 4, word 0, bit 0), programmed at factory test per the OEM's order.

| Mode | When used | OS-side consequence |
|---|---|---|
| DT | Linux direct boot; Android boot | RISC-V Linux reads FDT from `a1`; Android reads FDT from `a1`; no ACPI tables loaded |
| ACPI | Windows-via-Wine under Linux; NUMA/CXL-aware server Linux | RISC-V Linux reads ACPI tables via UEFI system table or via RSDP pointed to by a special DT node; OS uses ACPI for enumeration |

Default: DT. ACPI mode is opt-in for server-class configurations where NUMA and CXL topology need OS-level enumeration.

## .33 Device Tree Bindings

The DT is passed to the OS as a Flattened Device Tree (FDT) blob. The blob is generated by the boot ROM from a static DTS template compiled at build time, patched with per-product parameters (memory size, sensor count, crossbar presence, CXL presence) that are read from OTP.

### .33.1 Vendor prefix and naming

Custom compatible strings use the vendor prefix `ferrohybrid,` (lowercase, no capital, no space). The prefix is a descriptive namespace, not a company name. Standard RISC-V bindings (`riscv,`, `sifive,`, `riscv,plic0`, etc.) are used for standard components.

### .33.2 Root node and chosen

```
/ {
    #address-cells = <2>;
    #size-cells = <2>;
    model = "Ferroelectric Hybrid Processor Family";
    compatible = "ferrohybrid,pc-1", "ferrohybrid,fhpf";
    // patched at build time to one of:
    //   "ferrohybrid,ph-1", "ferrohybrid,pc-1", "ferrohybrid,dc-1s"

    chosen {
        bootargs = "console=ttySIF0,115200 earlycon=sbi";
        stdout-path = "/soc/serial@40005000";
        riscv,isa-base = "rv64";
        riscv,isa-extensions = "i,m,a,f,d,c,g,v,h";
    };
```

```
    cpus {
        #address-cells = <1>;
        #size-cells = <0>;
        // populated per product (see cpus node below)
    };

    memory@80000000 {
        device_type = "memory";
        reg = <0x0 0x80000000 0x0 0x0>;  // size patched per product
    };

    soc {
        #address-cells = <2>;
        #size-cells = <2>;
        compatible = "simple-bus";
        ranges;

        // all peripheral nodes below
    };
};
```

### .33.3 CPU nodes

The `cpus` node enumerates every hart. Ph-1 has 10 harts (6 big + 4 little); Pc-1 has 28 harts; Dc-1S has 32 harts. Each hart gets a `cpu@<hart-id>` node with a `cpu-map` entry.

```
cpus {
    #address-cells = <1>;
    #size-cells = <0>;

    cpu@0 {
        device_type = "cpu";
        compatible = "riscv";
        reg = <0>;
        status = "okay";
        riscv,isa = "rv64imafdcvh";
        mmu-type = "riscv,sv39";
        clock-frequency = <6200000000>;  // patched per product / bin
        cpu-idle-states = <&CPU_SLEEP_0 &CPU_SLEEP_1>;

        interrupt-controller {
            #interrupt-cells = <1>;
            compatible = "riscv,cpu-intc";
            interrupt-controller;
        };
    };

    // ... one cpu@N per hart, up to the product's hart count

    cpu-map {
        cluster0 {
            core0 { cpu = <&cpu0>; };
```

```
            core1 { cpu = <&cpu1>; };
            // ...
        };
        // Ph-1 has two clusters: cluster0 (big) and cluster1 (little)
    };

    idle-states {
        CPU_SLEEP_0: cpu-sleep-0 {
            compatible = "riscv,idle-state";
            riscv,sbi-sleep-type = <1>;  // WFI
            entry-latency-us = <10>;
            exit-latency-us = <10>;
            min-residency-us = <100>;
        };
        CPU_SLEEP_1: cpu-sleep-1 {
            compatible = "riscv,idle-state";
            riscv,sbi-sleep-type = <2>;  // P3 deep sleep
            entry-latency-us = <10>;
            exit-latency-us = <10000>;   // 10 µs wake
            min-residency-us = <1000000>;
        };
    };
};
```

### .33.4 Interrupt controller (PLIC-lite)

The PLIC-lite is a variant of the standard RISC-V PLIC (16 or 64 sources; single hart). The DT node declares it with the standard `riscv,plic0` compatible string plus a design-specific qualifier.

```
plic: interrupt-controller@40002000 {
    compatible = "ferrohybrid,plic-lite", "riscv,plic0";
    reg = <0x0 0x40002000 0x0 0x1000>;
    #interrupt-cells = <1>;
    interrupt-controller;
    riscv,ndev = <64>;         // 16 on Ph-1
    interrupts-extended = <&cpu0_intc 11>;
    // The single hart that services all PLIC interrupts
};
```

### .33.5 CLINT

Standard RISC-V CLINT at the standard address.

```
clint: timer@40001000 {
    compatible = "riscv,clint0";
    reg = <0x0 0x40001000 0x0 0x10000>;
    interrupts-extended = <&cpu0_intc 3 &cpu0_intc 7>;
};
```

### .33.6 Serial console

The host interface block (Appendix A §A.8) exposes a UART-like console on Ph-1. On Pc-1 and Dc-1S, the console is provided by the SoC's integrated UART (a standard 16550-compatible block added at 0x40005000 for this design).

```
serial@40005000 {
    compatible = "ns16550a";
    reg = <0x0 0x40005000 0x0 0x100>;
    clock-frequency = <1843200>;
    current-speed = <115200>;
    reg-shift = <2>;
    reg-io-width = <4>;
    interrupts-extended = <&plic 14>;
};
```

### .33.7 GPIO

```
gpio@40000000 {
    compatible = "ferrohybrid,gpio";
    reg = <0x0 0x40000000 0x0 0x100>;
    gpio-controller;
    #gpio-cells = <2>;
    interrupt-controller;
    #interrupt-cells = <2>;
    interrupts-extended = <&plic 20>;
    ngpios = <32>;              // 8 on Ph-1
    ferrohybrid,drive-strengths = <2 4 8 12>;  // mA
    ferrohybrid,pull-modes = "none", "up", "down", "keeper";
};
```

### .33.8 Thermal sensors

Each sensor is a separate node; the number of nodes is 4 (Ph-1), 6 (Pc-1), or 8 (Dc-1S).

```
thermal@40004000 {
    compatible = "ferrohybrid,thermal";
    reg = <0x0 0x40004000 0x0 0x100>;
    #thermal-sensor-cells = <1>;
    interrupts-extended = <&plic 10 &plic 11>;
};

thermal-zones {
    cpu-thermal0 {
        polling-delay-passive = <100>;
        polling-delay = <1000>;
        thermal-sensors = <&thermal 0>;
        trips {
            cpu_alert: cpu-alert {
                temperature = <85000>;   // 85°C in millidegrees
                hysteresis = <5000>;
                type = "passive";
            };
```

```
            cpu_crit: cpu-crit {
                temperature = <115000>;  // 115°C in millidegrees
                hysteresis = <0>;
                type = "critical";
            };
        };
        cooling-maps {
            map0 { trip = <&cpu_alert>; cooling-device = <&cpu0 1 3>; };
        };
    };
    // ... additional zones per sensor
};
```

### .33.9 NVL3 controller

```
nvl3: nvl3-controller@4000D000 {
    compatible = "ferrohybrid,nvl3-controller";
    reg = <0x0 0x4000D000 0x0 0x100>;
    interrupts-extended = <&plic 4 &plic 5 &plic 6 &plic 7 &plic 8>;
    // Single-bit ECC, double-bit ECC, verify, endurance, bad-block
    ferrohybrid,nvl3-size-mb = <64>;        // 24 for Dc-1S
    ferrohybrid,nvl3-banks = <256>;         // 96 for Dc-1S
    ferrohybrid,nvl3-window = <0x0 0x80000000 0x0 0x4000000>;  // 64 MB
    // ... additional properties for write-verify, pre-wake-up, endurance
};
```

### .33.10 Crossbar controller (Ph-1, Pc-1 only)

```
crossbar: crossbar-controller@4000E000 {
    compatible = "ferrohybrid,crossbar-controller";
    reg = <0x0 0x4000E000 0x0 0x100>;
    interrupts-extended = <&plic 16 &plic 17 &plic 18>;
    // MAC done, weight update, calibration
    ferrohybrid,crossbar-rows = <1024>;     // 512 on Ph-1
    ferrohybrid,crossbar-cols = <1024>;
    ferrohybrid,crossbar-bits = <2>;        // 2-bit baseline
    ferrohybrid,crossbar-mode = "parallel"; // "time-mux" on Ph-1
    ferrohybrid,custom-instructions = "0x0B";  // opcode space
};
```

### .33.11 Security engine and debug

```
security@4000A000 {
    compatible = "ferrohybrid,security-engine";
    reg = <0x0 0x4000A000 0x0 0x200>;
    interrupts-extended = <&plic 21 &plic 22 &plic 23 &plic 24 &plic 25>;
    // AES done, SHA done, ECDSA done, TRNG health, DICE
};

debug@4000B000 {
    compatible = "ferrohybrid,debug-auth";
    reg = <0x0 0x4000B000 0x0 0x100>;
```

```
    interrupts-extended = <&plic 15>;
};

otp@40008000 {
    compatible = "ferrohybrid,otp-controller";
    reg = <0x0 0x40008000 0x0 0x100>;
    interrupts-extended = <&plic 26 &plic 27>;
    // read complete, program failed
    ferrohybrid,word-size = <4>;            // 32-bit words
    ferrohybrid,total-bytes = <1048576>;    // 1 MB
};
```

### .33.12 DMA, vector block, power, watchdog

```
dma@40006000 {
    compatible = "ferrohybrid,dma";
    reg = <0x0 0x40006000 0x0 0x100>;
    interrupts-extended = <&plic 13>;
    dma-channels = <1>;
    #dma-cells = <1>;
    ferrohybrid,chunk-size = <4096>;
};

vector@40007000 {
    compatible = "ferrohybrid,vector-block";
    reg = <0x0 0x40007000 0x0 0x100>;
    interrupts-extended = <&plic 19>;
    status = "okay";                        // "disabled" on Dc-1S
};

power@40009000 {
    compatible = "ferrohybrid,power-mgmt";
    reg = <0x0 0x40009000 0x0 0x100>;
    interrupts-extended = <&plic 12>;
    ferrohybrid,dvfs-p-states = <5>;
    ferrohybrid,rails = "core", "io", "analog", "pll", "mem", "aux";
};

watchdog@40003000 {
    compatible = "ferrohybrid,watchdog";
    reg = <0x0 0x40003000 0x0 0x100>;
    interrupts-extended = <&plic 9>;
    ferrohybrid,timeout-modes = <1 10 100 1000>;  // ms
    ferrohybrid,reset-enable;
};
```

### .33.13 CXL (Pc-1, Dc-1S only)

```
cxl@4000C000 {
    compatible = "ferrohybrid,cxl-control";
    reg = <0x0 0x4000C000 0x0 0x100>;
    interrupts-extended = <&plic 28 &plic 29 &plic 30>;
```

```
    // link up, link down, link error
    ferrohybrid,cxl-version = "3.0";
    ferrohybrid,link-width = <16>;
    ferrohybrid,link-speed = <64>;         // GT/s
    status = "okay";                        // "disabled" on Ph-1
};
```

### .33.14 Chosen node (boot handoff)

The `chosen` node carries the boot arguments and the initrd location (if any), plus a design-specific property pointing at the runtime firmware services.

```
chosen {
    bootargs = "console=ttySIF0,115200 earlycon=sbi";
    stdout-path = "/soc/serial@40005000";
    linux,initrd-start = <0x0 0x0>;        // patched if initrd present
    linux,initrd-end = <0x0 0x0>;
    ferrohybrid,firmware-version = "1.0.0";
    ferrohybrid,boot-mode = "dt";           // or "acpi"
};
```

## .34 ACPI Tables

ACPI mode presents the platform via ACPI tables. The boot ROM generates the tables at build time and places them in reserved memory at a fixed address. The OS discovers them via the UEFI system table (if UEFI boot) or via the RSDP pointer in the DT `chosen` node (the DT is still passed in ACPI mode for the initial handoff, but the OS ignores the peripheral nodes and uses ACPI instead).

### .34.1 Table inventory

| Table | Purpose | Products | ACPI reference |
|---|---|---|---|
| RSDP | Root pointer (points to XSDT) | All | ACPI 6.5 §5.2.5 |
| XSDT | Extended System Description Table | All | ACPI 6.5 §5.2.8 |
| FADT | Fixed ACPI Description Table | All | ACPI 6.5 §5.2.9 |
| DSDT | Differentiated System Description Table | All | ACPI 6.5 §5.2.11 |
| MADT | Multiple APIC Description Table (RISC-V) | All | ACPI 6.5 §5.2.12 |
| GTDT | Generic Timer Description Table | All | ACPI 6.5 §5.2.24 |
| RHCT | RISC-V Hart Capabilities Table | All | ACPI 6.5 §5.2.37 |
| SPCR | Serial Port Console Redirection | All | Microsoft SPCR spec |
| SRAT | System Resource Affinity Table (NUMA) | All | ACPI 6.5 §5.2.16 |

| Table | Purpose | Products | ACPI reference |
|---|---|---|---|
| SLIT | System Locality Information Table | All | ACPI 6.5 §5.2.17 |
| PPTT | Processor Properties Topology Table | All | ACPI 6.5 §5.2.30 |
| MCFG | Memory Mapped Configuration space | Pc-1, Dc-1S | PCI Firmware spec |
| IORT | I/O Remapping Table | Pc-1, Dc-1S | ACPI 6.5 §5.2.33 |
| CEDT | CXL Early Discovery Table | Pc-1, Dc-1S | CXL 3.0 spec |

### .34.2 FADT

Follows ACPI 6.5 §5.2.9. The design-specific values:

| Field | Ph-1 | Pc-1 | Dc-1S |
|---|---|---|---|
| Preferred PM profile | Mobile | Desktop | Enterprise Server |
| SCI interrupt | Source 12 (brown-out) | Source 12 | Source 12 |
| SCI interrupt mode | level | level | level |
| PM1a_EVT_BLK | OTP region 4 | OTP region 4 | OTP region 4 |
| PM1_CNT_LEN | 2 | 2 | 2 |
| FLAGS.CENTURY | 0 | 0 | 0 |
| FLAGS.RTC_S4 | 0 | 0 | 0 |
| FLAGS.RESET_REG_SUP | 1 | 1 | 1 |
| RESET_REG | 0x40009014 | 0x40009014 | 0x40009014 |
| RESET_VALUE | 0xDEAD | 0xDEAD | 0xDEAD |
| ARM_BOOT_ARCH | 0 | 0 | 0 |

### .34.3 MADT (RISC-V variant)

Follows ACPI 6.5 §5.2.12. Contains the following RISC-V-specific entries:

- **Type 0x18 (RISC-V Hart)**: one entry per hart. `ACPI Processor ID` = hart ID; `Hart ID` = hart ID; `Flags` bit 0 = 1 (hart enabled); `ACPI Processor UID` = hart ID.
- **Type 0x1B (RISC-V PLIC)**: one entry for the PLIC-lite. `RISC-V PLIC ID` = 0; `Hardware ID` = 0x40002000; `Platform Interrupt Sources` = 64 (16 on Ph-1); `GSI Base` = 0; `Maximum Priority` = 7.
- **Type 0x1C (RISC-V APLIC)**: not used in this design (single-hart PLIC; no MSI).
- **Type 0x0D (RISC-V IMSIC)**: not used.
- **Type 0x19 (RISC-V IMSIC per hart)**: not used.

### .34.4 GTDT

Follows ACPI 6.5 §5.2.24. The generic timer comes from the CLINT (standard RISC-V mtime/mtimecmp). `CntControlBase` = 0x40001000 + 0x4000 (standard CLINT mtimecmp offset); `CntReadBase` = 0x40001000 + 0xBFF8 (standard CLINT mtime offset).

### .34.5 RHCT

Follows ACPI 6.5 §5.2.37. Lists the RISC-V ISA extensions supported by every hart: i, m, a, f, d, c, g, v, h, zicsr, zifencei, zicntr, zihpm.

### .34.6 SRAT and SLIT (NUMA)

SRAT follows ACPI 6.5 §5.2.16. Contains:

- **Processor Local APIC/SAPIC Affinity**: one entry per hart, `Proximity Domain` = the NUMA node that contains the hart.
- **Memory Affinity**: one entry per contiguous memory range, `Proximity Domain` = the NUMA node that contains that memory.
- **Generic Initiator Affinity**: one entry per CXL device (Pc-1, Dc-1S), `Proximity Domain` = the NUMA node closest to that CXL device.

Per-product NUMA topology:

| Product | NUMA nodes | Contents |
|---|---|---|
| Ph-1 | 1 | All harts + all LPDDR5X |
| Pc-1 | 1 (single-socket) or 2 (2-socket via CXL) | Node 0 = local harts + local DDR5 + local NVL3; Node 1 = remote socket's harts + memory + NVL3 (via CXL) |
| Dc-1S | 1 (single-socket) or 2–4 (multi-socket via CXL) | Node 0 = local harts + local DDR5 + local NVL3; Nodes 1..N = remote sockets |

SLIT follows ACPI 6.5 §5.2.17. Distances: 10 (local) on the diagonal; 20 for 1-hop CXL (adjacent socket); 30 for 2-hop CXL (ring topology on multi-socket Dc); 40 for more distant paths (if any).

### .34.7 MCFG, IORT, CEDT

MCFG (Pc-1, Dc-1S): PCIe ECAM base at $\texttt{0x0000}_0003_0000_0000$; $16 busesperrootcomplex$; $basebus = 0$; $segment = 0$.

`IORT (Pc-1, Dc-1S): one ITS-free IORT (RISC-V does not have a GIC ITS); one SMMU-v3 node per PCIe root complex; one RMR (Root Memory Region) per PCIe device for MSI doorbells.`

`CEDT (Pc-1, Dc-1S): follows CXL 3.0 §9.18. Contains:`

- `CHBS (CXL Host Bridge Structure): one entry per CXL host bridge; Base HPA = the CXL.mem target address of the host bridge.`
- `CFMWS (CXL Fixed Memory Window Structure): one entry per memory window; describes the range of host physical addresses that map to CXL devices.`

### .34.8 PPTT

Follows ACPI 6.5 §5.2.30. Describes the CPU topology (packages, clusters, cores). One socket per physical package; one node per cluster (4--6 per die); one node per core.

## .35 DSDT Namespace

The DSDT declares the custom devices in the ACPI namespace using ASL. Every device gets a _HID (Hardware ID) that is a design-specific string, a _CRS (Current Resource Settings) descriptor that describes its MMIO range and interrupt, and _DSD (Device-Specific Data) properties that describe the design-specific parameters.

### .35.1 Vendor ID

The design-specific _HID prefix is FHP (Ferroelectric Hybrid Processor). The full _HID for each device is FHP0001 through FHP000E (12 custom devices, as follows):

| _HID | Device | Compatible DT node (for reference) |
|---|---|---|
| FHP0001 | GPIO | ferrohybrid,gpio |
| FHP0002 | PLIC-lite | ferrohybrid,plic-lite |
| FHP0003 | Watchdog | ferrohybrid,watchdog |
| FHP0004 | Thermal | ferrohybrid,thermal |
| FHP0005 | Host interface | ns16550a (serial) or PCIe config |
| FHP0006 | DMA | ferrohybrid,dma |
| FHP0007 | Vector block | ferrohybrid,vector-block |
| FHP0008 | OTP | ferrohybrid,otp-controller |
| FHP0009 | Power management | ferrohybrid,power-mgmt |
| FHP000A | Security engine | ferrohybrid,security-engine |
| FHP000B | Debug auth | ferrohybrid,debug-auth |
| FHP000C | CXL control | ferrohybrid,cxl-control |
| FHP000D | NVL3 controller | ferrohybrid,nvl3-controller |
| FHP000E | Crossbar controller | ferrohybrid,crossbar-controller |

### .35.2 DSDT source (ASL)

The full DSDT for the Pc-1 product. The Ph-1 and Dc-1S versions differ only in the values of the numeric parameters and the presence of the crossbar (Ph-1) or CXL (Dc-1S). The differences are noted inline.

```
DefinitionBlock ("", "DSDT", 2, "FHP   ", "FHPPC1  ", 0x00000001)
{
    Scope (\_SB)
    {
        Device (GPIO) {
            Name (_HID, "FHP0001")
            Name (_UID, 0)
```

```
    Name (_CRS, ResourceTemplate () {
        Memory32Fixed (ReadWrite, 0x40000000, 0x100)
        Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 20 }
    })
    Name (_DSD, Package () {
        ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
        Package () {
            Package () { "ngpios", 32 },
            Package () { "drive-strengths", Package () { 2, 4, 8, 12 } },
            Package () { "pull-modes", Package () { "none", "up", "down", "keeper" }
        }
    })
    Method (_STA) { Return (0x0F) }
}

Device (PLIC) {
    Name (_HID, "FHP0002")
    Name (_UID, 0)
    Name (_CRS, ResourceTemplate () {
        Memory32Fixed (ReadWrite, 0x40002000, 0x1000)
    })
    Name (_DSD, Package () {
        ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
        Package () {
            Package () { "riscv,ndev", 64 },
            Package () { "interrupts-extended", Package () { 0, 11 } }
        }
    })
    Method (_STA) { Return (0x0F) }
}

Device (WDT0) {
    Name (_HID, "FHP0003")
    Name (_UID, 0)
    Name (_CRS, ResourceTemplate () {
        Memory32Fixed (ReadWrite, 0x40003000, 0x100)
        Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 9 }
    })
    Name (_DSD, Package () {
        ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
        Package () {
            Package () { "timeout-modes-ms", Package () { 1, 10, 100, 1000 } },
            Package () { "reset-enable", 1 }
        }
    })
    Method (_STA) { Return (0x0F) }
}

Device (TEMP) {
    Name (_HID, "FHP0004")
    Name (_UID, 0)
```

```
    Name (_CRS, ResourceTemplate () {
        Memory32Fixed (ReadWrite, 0x40004000, 0x600)   // 6 sensors × 0x100
        Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 10, 11 }
    })
    Name (_DSD, Package () {
        ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
        Package () {
            Package () { "num-sensors", 6 },
            Package () { "sensor-offset", 0x100 }
        }
    })
    Method (_STA) { Return (0x0F) }
}

Device (NVL0) {
    Name (_HID, "FHP000D")
    Name (_UID, 0)
    Name (_CRS, ResourceTemplate () {
        Memory32Fixed (ReadWrite, 0x4000D000, 0x100)
        Memory32Fixed (ReadWrite, 0x80000000, 0x4000000)   // 64 MB NVL3 window
        Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 4, 5, 6, 7, 8 }
    })
    Name (_DSD, Package () {
        ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
        Package () {
            Package () { "nvl3-size-mb", 64 },
            Package () { "nvl3-banks", 256 },
            Package () { "nvl3-endurance-write-count", 215 },
            Package () { "nvl3-retention-years", 10 }
        }
    })
    Method (_STA) { Return (0x0F) }
}

Device (XBAR) {
    Name (_HID, "FHP000E")
    Name (_UID, 0)
    Name (_CRS, ResourceTemplate () {
        Memory32Fixed (ReadWrite, 0x4000E000, 0x100)
        Interrupt (ResourceConsumer, Level, ActiveHigh, Exclusive) { 16, 17, 18 }
    })
    Name (_DSD, Package () {
        ToUUID("daffd814-6eba-4d8c-8a91-bc9bbf4aa301"),
        Package () {
            Package () { "crossbar-rows", 1024 },
            Package () { "crossbar-cols", 1024 },
            Package () { "crossbar-bits", 2 },
            Package () { "crossbar-mode", "parallel" }
        }
    })
    Method (_STA) { Return (0x0F) }
```

```
        }

        // ... additional devices (DMA, Vector, OTP, Power, Security, Debug, CXL)
        // follow the same structure; see the table above for _HID and resource
        // definitions
    }

    // Thermal zones declared at the root scope
    Scope (\_TZ)
    {
        ThermalZone (TZ00) {
            Method (_TMP, 0, Serialized) {
                Return (\_SB.TEMP.TEMP0)      // read sensor 0
            }
            Method (_CRT, 0, Serialized) { Return (388) }   // 115°C in 0.1K
            Method (_PSV, 0, Serialized) { Return (358) }   // 85°C in 0.1K
            Name (_TC1, 2)
            Name (_TC2, 5)
            Name (_TSP, 10)
        }
        // additional zones per sensor
    }

    // Processor objects (one per hart, matching MADT Type 0x18)
    Scope (\_SB)
    {
        Processor (CPU0, 0, 0x00000410, 0x06) {}
        Processor (CPU1, 1, 0x00000410, 0x06) {}
        // ... one per hart, up to the product's hart count
    }
}
```

The crossbar XBAR device is only present on Ph-1 and Pc-1. The _STA method on Dc-1S returns 0 (device not present).

The CXL device is only present on Pc-1 and Dc-1S; its _CRS declares the link control block at 0x4000C000 and the interrupt sources 28--30.

## .36 Boot Handoff Protocol

### .36.1 DT mode

The boot ROM, after completing the firmware sequence (Appendix G), passes control to the OS or bootloader with:

- a0 = boot hart ID (must be 0, the boot hart).
- a1 = physical address of the FDT blob in memory.
- pc = entry point of the kernel or bootloader.
- mstatus.MIE = 0 (interrupts disabled).
- satp = 0 (paging off; kernel sets up its own).

- All PLIC sources disabled; firmware does not leave any interrupt pending.
- The FDT is at a 2 MB-aligned address in a memory region not claimed by the kernel.

### .36.2 ACPI mode

The boot ROM:

- Generates the ACPI tables at build time into a reserved memory region (typically 4 MB at the top of DDR5).
- Sets up the RSDP pointer in the DT chosen node ferrohybrid,acpi-rsdp property.
- Passes a0 = 0, a1 = FDT address (the DT still enumerates the boot hart and memory; the OS uses the RSDP to find the full ACPI tables and ignores the DT peripheral nodes).
- The kernel detects the RSDP and switches to ACPI-based enumeration.

The ferrohybrid,acpi-rsdp DT property is a standard UINT64 in the chosen node; the OS-side kernel patch reads it and calls acpi_early_init with the RSDP address.

### .36.3 Per-product differences

| Parameter | Ph-1 | Pc-1 | Dc-1S |
|---|---|---|---|
| Default boot mode | DT | ACPI | ACPI |
| Hart count (MADT/DT) | 10 | 28 | 32 |
| NUMA nodes (SRAT) | 1 | 1--2 | 1--4 |
| CXL presence (CEDT) | no | yes | yes |
| Crossbar presence (XBAR) | yes | yes | no |
| DDR5 base | 0x0_8000_0000 | 0x0_8000_0000 | 0x0_8000_0000 |
| DDR5 size | patched per SKU | patched per SKU | patched per SKU |
| SPCR baud | 115200 | 115200 | 115200 |

## .37 Validation and Test

### .37.1 ACPI validation (pre-tape-out)

The ACPI tables are generated by a build script from a machine-readable description (typically a YAML file per product). Before tape-out, the tables are validated:

1. ACPI CA checker: build the tables with the ACPI Component Architecture (ACPICA) compiler, run the acpixtract -a extractor, and run the ACPICA acpiexec validator. All tables must pass with zero errors and zero warnings except for documented product-specific allowances.

2. Linux ACPI parser dry run: run the tables through the Linux ACPI parser on a host x86 machine (the parser is architecture-independent; the OS will reject the tables on x86 because the MADT has RISC-V entries, but the parse itself must succeed without errors before the OS examines the architecture).

3. Windows ACPI checker: for OEM partners who will support Windows-on-Wine, run Microsoft's ACPI validation tool (ASL compiler and firmware test suite).

4. Manual review: the DSDT namespace is reviewed by hand against the register map in Appendix A, to confirm that every $_{C}RSrangeand_{D}SDpropertymatchesthehardware.$

### .37.2 DT validation (pre-tape-out)

The DT nodes are validated:

1. dtc (device tree compiler): compile the DTS to DTB; require zero warnings from dtc -W all.

2. dt-schema: run the DT through the dt-schema validator (the Linux DT schema repository), which checks that every node's compatible string matches a known schema. Custom compatible strings (ferrohybrid,*) use the design-supplied schema files, which the design team maintains alongside the DTS.

3. Linux DT parser dry run: boot a Linux kernel with earlycon on QEMU using the DTB; the kernel must accept the DT without errors up to the point where real hardware is needed.

### .37.3 Hardware bring-up test (post-silicon)

Once the first silicon returns, the boot ROM is patched with the real OTP values and the test sequence is:

1. Boot into a bare-metal firmware image that reads the DTB/ACPI tables and prints their parsed content to the UART.

2. Verify every node / table entry against the register map. Discrepancy stops the bring-up.

3. Boot into a real Linux kernel with the DT (DT mode). The kernel must reach userspace.

4. Boot into a real Linux kernel with the ACPI tables (ACPI mode). The kernel must reach userspace and dmesg | grep -i acpi must show the expected enumeration.

5. For server products, boot into a NUMA-enabled kernel and confirm numactl --hardware matches the expected topology.

### .37.4 Common failures

| Symptom | Common cause |
|---|---|
| Kernel panics at ACPI parse | RSDP checksum mismatch; check the _CRS checksum in the DSDT |
| Kernel finds only 1 CPU | MADT Type 0x18 flags bit 0 is 0 for other harts; or DT status is "disabled" |
| NUMA topology empty | SRAT/SLIT missing or SRAT Proximity Domain values do not match SLIT |
| CXL not enumerated | CEDT missing or CHBS.Base HPA does not match the memory window |
| NVL3 not detected | DSDT NVL0 $_{C}RSwindowdoesnotmatchtheregistermap, orDT$nvl3-window$is$ |
| Crossbar driver probe fails | XBAR $_{H}IDmissingorcustom - instructionopcodemismatch$ |
| Serial console dead | SPCR baud rate or SPCR Base Address mismatch |
| Thermal zone does not throttle | _TZ.$_{T}MPreturnsaconstantorthesensorindexisoutofrange$ |

The single most common bring-up error is a mismatch between the DSDT $_{C}RSrangeandtheactua$

## .38 Summary

| Artifact | Delivered with | Notes |
|---|---|---|
| DTS template | this appendix (structure) + design repo (full file) | Compiled at build time |
| ACPI tables | generated from product YAML at build time | DSDT/SSDT compiled via ACPICA |
| DSDT source (ASL) | this appendix (full source) | Hand-maintained per product |
| Product YAML | design repo | One per product |
| ACPI validation report | pre-tape-out | ACPICA + Windows checker |
| DT validation report | pre-tape-out | dtc + dt-schema |

The DSDT source in §D.4 is the authoritative design definition for ACPI mode. The DT node structure in §D.3 is the authoritative design definition for DT mode. Any firmware that boots the design in either mode must generate tables that conform to the corresponding structure in this appendix.

# Appendix E: Full Pin Description

This appendix specifies the ball assignment, signal catalog, differential pair catalog, power/ground ball assignment, and critical signal placement for the three packages used in the three Tier-1 products. The complete per-ball mapping is provided as a companion CSV file (ballmap_<product>.csv) that ships with this specification; this appendix gives the format specification, the signal catalog, and the placement rules needed to validate or regenerate that file.

## .39 Ballmap CSV File Format

Each product ships with a ballmap CSV file. The file is the authoritative per-ball assignment; this appendix defines its format and the design rules that generate it.

### .39.1 File naming

ballmap_ph1.csv, ballmap_pc1.csv, ballmap_dc1s.csv.

### .39.2 Column definitions

The CSV has exactly these columns, in this order:

| Column | Type | Example | Meaning |
|---|---|---|---|
| Ball | string | A1 | Ball coordinate (row letter + column number) |
| NetName | string | VDD_CORE | Net name as used on the board |
| SignalGroup | enum | POWER_CORE | Signal group name from the catalog in §E.3 |
| Rail | enum | VDD_CORE | Power rail (blank for signals) |
| Direction | enum | POWER | In, Out, Bidir, NC, POWER, GND |
| IOStandard | string | 1.0V_CMOS | I/O standard |
| DiffPair | string | PCIe_TX0 | Differential pair name (blank if single-ended) |
| Polarity | char | P or N | P or N (blank if single-ended) |
| Function | string | PCIe Gen5 TX lane 0 | Functional description |

A sample of the first five rows (one per category):

```
Ball,NetName,SignalGroup,Rail,Direction,IOStandard,DiffPair,Polarity,Function
A1,VDD_CORE,VDD_CORE,VDD_CORE,POWER,1.0V,POWER,,,
A2,VSS,VSS,VSS,POWER,0V,GND,,,
A3,PCIe_TX0_P,PCIE,PCIe,Out,PCIe5,PCIE_TX0,P,PCIe Gen5 TX lane 0 positive
A4,PCIe_TX0_N,PCIE,PCIe,Out,PCIe5,PCIE_TX0,N,PCIe Gen5 TX lane 0 negative
A5,DDR5_CH0_DQ0,DDR5,DDR5,Bidir,DDR5_0V,DDR5_CH0_DQ0,P,DDR5 channel 0 DQ bit 0
```

### .39.3 Validation rules

Every ballmap CSV must satisfy:

1. Every ball coordinate in the package grid appears exactly once.
2. Every NetName appears on exactly the number of balls required by the catalog (§E.3).
3. Every DiffPair appears on exactly two balls (one P, one N).
4. Ball coordinate rows and columns must be contiguous (no gaps) except where the package depopulates corners or power/ground regions.

The validation script (ballmap_validate.py) is provided with the specification. It checks these rules and produces a report.

## .40 Signal Catalog

This section lists every signal group used in the three products, with ball counts. The catalog is the source of truth for the ballmap CSV: every ball belongs to exactly one signal group.

### .40.1 Ph-1 (FCBGA 400-ball)

| Signal Group | Balls | Rail/Standard | Notes |
|---|---|---|---|
| VDD_CORE | 40 | 1.0V | --- |
| VDD_IO | 20 | 1.8V | --- |
| VDD_ANALOG | 20 | 1.5V | Crossbar, NVL3 readout |
| VDD_PLL | 4 | 1.0V | --- |
| VSS | 80 | 0V | --- |
| LPDDR5X_CH0 | 40 | LPDDR5X | Channel 0 (32 DQ + 4 DQS/CK/CA pairs) |
| LPDDR5X_CH1 | 40 | LPDDR5X | Channel 1 |
| PCIE | 8 | PCIe3 | 4 diff pairs (2 TX + 2 RX) |
| USB | 4 | USB2 | D+, D-, VBUS, GND_ref |
| CAMERA_MIPI | 16 | MIPI_DPHY | 4-lane D-PHY (8 diff pairs) |
| DISPLAY_MIPI | 8 | MIPI_DPHY | 4-lane DSI (4 diff pairs) |

| Signal Group | Balls | Rail/Standard | Notes |
|---|---|---|---|
| MODEM_5G | 16 | MIPI_MPHY | 4-lane M-PHY (8 diff pairs) |
| SECURITY_DEBUG | 16 | 1.8V_CMOS | JTAG + secure debug |
| CONTROL | 8 | 1.8V_CMOS | SPI, SMBus, reset, interrupt |
| THERMAL | 2 | analog | Thermal diode |
| Total | 400 | | |

### .40.2 Pc-1 (LGA 1851-ball)

| Signal Group | Balls | Rail/Standard | Notes |
|---|---|---|---|
| VDD_CORE | 400 | 1.0--1.05V | --- |
| VDD_IO | 60 | 1.8V | --- |
| VDD_ANALOG | 100 | 1.5V | Crossbar, NVL3 readout |
| VDD_PLL | 8 | 1.0V | --- |
| VSS | 600 | 0V | --- |
| DDR5_CH0..7 | 320 | DDR5 | 8 channels × 40 balls (32 DQ + 4 DQS/CK/CA pairs) |
| PCIE | 32 | PCIe5 | 16 diff pairs (8 TX + 8 RX lanes) |
| CXL | 32 | PCIe5/CXL | 16 diff pairs (8 TX + 8 RX lanes) |
| SPI_JTAG_SMBUS | 20 | 1.8V_CMOS | --- |
| CONTROL | 28 | 1.8V_CMOS | reset, interrupt, BMC interface, etc. |
| SPARE | 251 | GND | Grounded spare |
| Total | 1851 | | |

### .40.3 Dc-1S (LGA 4677-ball)

| Signal Group | Balls | Rail/Standard | Notes |
|---|---|---|---|
| VDD_CORE | 900 | 1.0--1.05V | --- |
| VDD_IO | 100 | 1.8V | --- |
| VDD_ANALOG | 60 | 1.5V | NVL3 readout |
| VDD_PLL | 16 | 1.0V | --- |
| VSS | 1800 | 0V | --- |
| DDR5_CH0..3 | 160 | DDR5 | 4 channels × 40 balls |

| Signal Group | Balls | Rail/Standard | Notes |
|---|---|---|---|
| PCIE | 64 | PCIe5 | 32 diff pairs (16 TX + 16 RX lanes) |
| CXL | 32 | PCIe5/CXL | 16 diff pairs (8 TX + 8 RX lanes) |
| SPI_JTAG_SMBUS_BMC | 10 | 1.8V_CMOS | Includes BMC interface |
| CONTROL | 25 | 1.8V_CMOS | reset, interrupt, etc. |
| SPARE | 1420 | GND | Grounded spare |
| Total | 4677 | | |

## .41 Differential Pair Catalog

Every differential pair is listed with its net name and its pair members. The ballmap CSV assigns each pair member to a specific ball coordinate; this section lists the pairs themselves.

### .41.1 PCIe Gen5 (Pc-1, Dc-1S)

Standard 8-lane bidirectional PCIe Gen5. Each lane is one TX pair + one RX pair.

| Lane | TX pair (P, N) | RX pair (P, N) | Notes |
|---|---|---|---|
| 0 | PCIe_TX0_P, PCIe_TX0_N | PCIe_RX0_P, PCIe_RX0_N | Root complex port 0 |
| 1 | PCIe_TX1_P, PCIe_TX1_N | PCIe_RX1_P, PCIe_RX1_N | --- |
| 2 | PCIe_TX2_P, PCIe_TX2_N | PCIe_RX2_P, PCIe_RX2_N | --- |
| 3 | PCIe_TX3_P, PCIe_TX3_N | PCIe_RX3_P, PCIe_RX3_N | --- |
| 4 | PCIe_TX4_P, PCIe_TX4_N | PCIe_RX4_P, PCIe_RX4_N | --- |
| 5 | PCIe_TX5_P, PCIe_TX5_N | PCIe_RX5_P, PCIe_RX5_N | --- |
| 6 | PCIe_TX6_P, PCIe_TX6_N | PCIe_RX6_P, PCIe_RX6_N | --- |
| 7 | PCIe_TX7_P, PCIe_TX7_N | PCIe_RX7_P, PCIe_RX7_N | --- |

Total: 16 pairs, 32 balls. On Dc-1S, 16 PCIe lanes are provided (32 pairs, 64 balls); lanes 8--15 use the same naming convention with index 8--15.

### .41.2 CXL 3.0 (Pc-1, Dc-1S)

Standard 8-lane bidirectional CXL.

| Lane | TX pair (P, N) | RX pair (P, N) | Notes |
|---|---|---|---|
| 0 | CXL_TX0_P, CXL_TX0_N | CXL_RX0_P, CXL_RX0_N | --- |
| 1 | CXL_TX1_P, CXL_TX1_N | CXL_RX1_P, CXL_RX1_N | --- |
| 2 | CXL_TX2_P, CXL_TX2_N | CXL_RX2_P, CXL_RX2_N | --- |
| 3 | CXL_TX3_P, CXL_TX3_N | CXL_RX3_P, CXL_RX3_N | --- |
| 4 | CXL_TX4_P, CXL_TX4_N | CXL_RX4_P, CXL_RX4_N | --- |
| 5 | CXL_TX5_P, CXL_TX5_N | CXL_RX5_P, CXL_RX5_N | --- |
| 6 | CXL_TX6_P, CXL_TX6_N | CXL_RX6_P, CXL_RX6_N | --- |
| 7 | CXL_TX7_P, CXL_TX7_N | CXL_RX7_P, CXL_RX7_N | --- |

Total: 16 pairs, 32 balls. Dc-1S uses the same 8-lane CXL, giving 32 balls.

### .41.3 DDR5 (Pc-1, Dc-1S)

Per channel: 32 DQ (single-ended) + 4 DQS pairs (8 balls). 40 balls per channel. The DQS pairs are:

| Byte lane | DQS pair (P, N) | DQ bits |
|---|---|---|
| 0 | DDR5_CHn_DQS0_P, DDR5_CHn_DQS0_N | DQ0--DQ7 |
| 1 | DDR5_CHn_DQS1_P, DDR5_CHn_DQS1_N | DQ8--DQ15 |
| 2 | DDR5_CHn_DQS2_P, DDR5_CHn_DQS2_N | DQ16--DQ23 |
| 3 | DDR5_CHn_DQS3_P, DDR5_CHn_DQS3_N | DQ24--DQ31 |

The command/address (CA) and clock (CK) signals are handled by the DDR5 controller IP; they use single-ended CA in this design, not the differential CA of standard DDR5, to keep the ball count at 40 per channel. This is a deliberate ball-count optimization; the DDR5 controller IP supports both modes.

For Pc-1 (8 channels): 32 DQS pairs total. For Dc-1S (4 channels): 16 DQS pairs total.

### .41.4 LPDDR5X (Ph-1)

Per channel: 40 balls. Same arrangement as DDR5 | 32 DQ + 4 DQS pairs per channel = 40 balls.

2 channels: 64 DQ balls + 8 DQS pairs (16 balls) = 80 balls total, matches the 80-ball LPDDR5X group.

8 DQS pairs.

### .41.5 Camera MIPI (Ph-1)

4-lane MIPI D-PHY, 8 diff pairs = 16 balls.

Lanes 0--3, each lane = 1 differential pair. The 8 pairs are the 4 data lanes (RX from the camera) plus 4 additional lanes (or 2 clock pairs + 4 data, etc.; the exact arrangement is in the ballmap CSV).

### .41.6 Display MIPI (Ph-1)

8 balls = 4 diff pairs. 4-lane DSI: 4 data pairs (or 3 data + 1 clock).

### .41.7 5G Modem (Ph-1)

16 balls = 8 diff pairs. M-PHY (MIPI M-PHY) is bidirectional multi-lane. 4-lane × 2 directions = 8 pairs.

### .41.8 Summary of diff pairs by product

| Interface | Ph-1 pairs | Pc-1 pairs | Dc-1S pairs |
|---|---|---|---|
| PCIe | 4 | 16 | 32 |
| CXL | 0 | 16 | 16 |
| DDR5 | 0 | 32 | 16 |
| LPDDR5X | 8 | 0 | 0 |
| Camera MIPI | 8 | 0 | 0 |
| Display MIPI | 4 | 0 | 0 |
| 5G Modem | 8 | 0 | 0 |
| Total diff pairs | 32 | 64 | 64 |

## .42 Power and Ground Ball Assignment

Every power and ground ball is assigned to one rail (or VSS). The rails and their ball counts are listed in §E.3. The physical placement follows these rules:

### .42.1 Placement rules

1. VSS interleaving: VSS balls are interleaved with signal balls at a ratio of at least 1 VSS per 4 high-speed signal balls, and 1 VSS per 8 single-ended signal balls, to provide return-current paths.

2. Power/ground rings: VDD_CORE balls are concentrated in the center of the package (for lowest PDN impedance to the die power grid). VSS balls form an outer ring and an inner interleave.

3. Rail separation: VDD_IO, VDD_ANALOG, and VDD_PLL are in separate quadrants or contiguous blocks, physically separated from each other to prevent coupling.

4. Analog isolation: VDD_ANALOG (crossbar, NVL3 readout) and VDD_PLL are surrounded by VSS guard balls.

5. High-speed signals: PCIe, CXL, DDR5, and LPDDR5X signals are placed on the package edge closest to the corresponding PHY in the die floorplan.

6. Spare balls: Pc-1's 251 spare and Dc-1S's 1420 spare are all VSS-grounded.

### .42.2 VDD_CORE placement

VDD_CORE balls are physically clustered to form a low-inductance power delivery network. For Pc-1 (400 VDD_CORE balls) and Dc-1S (900 VDD_CORE balls), the balls form a contiguous block in the package center, with VSS interleaved at 1 VSS per 2 VDD_CORE in the interior and 1 VSS per 1 VDD_CORE at the boundary.

For Ph-1 (40 VDD_CORE balls), the VDD_CORE balls are arranged in a 5×8 grid in the package center.

### .42.3 VDD_ANALOG placement

VDD_ANALOG balls are placed on the quadrant adjacent to the analog PHY (crossbar and NVL3 readout) in the die floorplan. On Pc-1, this is the bottom-left quadrant; on Ph-1, the top-right; on Dc-1S, the bottom-right.

The VDD_ANALOG group is isolated by 2 VSS guard rings (one inner, one outer) to prevent noise coupling from the core and I/O rails.

### .42.4 VDD_PLL placement

VDD_PLL balls (4 on Ph-1, 8 on Pc-1, 16 on Dc-1S) are placed immediately adjacent to the analog PLL block in the die floorplan, with dedicated VSS return balls.

## .43 Critical Signal Placement

For high-speed signal routing, the ball coordinate regions are constrained as follows.

### .43.1 PCIe Gen5 (Pc-1, Dc-1S)

All PCIe Gen5 pairs are placed on the package bottom edge (rows near the edge, e.g., the last 4 rows). TX and RX pairs for a given lane are adjacent to each other (within 2 ball columns) to allow skew-matched routing on the board. Lane 0 is the leftmost, lane 7 (or 15) is the rightmost.

### .43.2 CXL 3.0 (Pc-1, Dc-1S)

All CXL pairs are placed on the package left edge, mirroring the PCIe layout. Lane 0 is the bottom-most, lane 7 is the top-most.

### .43.3 DDR5 (Pc-1, Dc-1S)

DDR5 channels are split between two edges:

- Channels 0--3: top edge, arranged left to right.
- Channels 4--7: right edge, arranged top to bottom (Pc-1 only; Dc-1S has 4 channels only, all on the top edge).

Within each channel, the DQ byte lanes are grouped, the DQS pair for each byte lane is adjacent to the corresponding DQ group, and the CA signals are grouped near the DQS of byte lane 0.

### .43.4 LPDDR5X (Ph-1)

LPDDR5X channels are on the top edge (channel 0) and right edge (channel 1), both near the corresponding PHY in the die floorplan.

### .43.5 MIPI (Ph-1)

Camera MIPI is on the top edge (near the camera ISP), display MIPI on the right edge (near the display controller), 5G modem on the bottom edge (near the modem interface).

### .43.6 Placement summary table

| Interface | Ph-1 edge | Pc-1 edge | Dc-1S edge |
|---|---|---|---|
| PCIe | bottom | bottom | bottom |
| CXL | (none) | left | left |
| DDR5 | (none) | top (0--3) + right (4--7) | top |
| LPDDR5X | top + right | (none) | (none) |
| Camera MIPI | top | (none) | (none) |
| Display MIPI | right | (none) | (none) |
| 5G modem | bottom | (none) | (none) |

## .44 No-Connect Balls

Balls that are not connected to any net inside the die are marked NC in the ballmap CSV, are physical balls on the package but are not bonded to any die pad, and are typically not connected on the board.

For the three packages:

| Package | NC balls | Location |
|---|---|---|
| Ph-1 | 0 | No NC balls (all 400 balls are assigned) |
| Pc-1 | 0 | No NC balls (all 1851 assigned; spare = 251 VSS) |
| Dc-1S | 0 | No NC balls (all 4677 assigned; spare = 1420 VSS) |

All spare balls in Pc-1 and Dc-1S are tied to VSS (not floated), to provide additional PDN return paths. The ballmap CSV marks them as "SPARE" signal group, "GND" rail, "POWER" direction.

## .45 Electrical Specification

Each signal group has an electrical specification (voltage, drive current, termination, differential impedance).

### .45.1 PCIe Gen5 (Pc-1, Dc-1S)

| Parameter | Value | Notes |
|---|---|---|
| TX differential swing | 800--1300 mV | Peak-to-peak, configurable in PHY |
| RX differential sensitivity | 25 mV | At BER $< 10^{-12}$ |
| Differential impedance | 85 $\Omega$ $\pm$10% | PCB trace design target |
| Common-mode impedance | 45--55 $\Omega$ | --- |
| AC coupling | 100 nF | TX side, per lane |
| Maximum trace length | 300 mm | On FR4 with suitable stackup |

### .45.2 CXL 3.0 (Pc-1, Dc-1S)

Same as PCIe Gen5 (CXL uses the same PHY layer).

### .45.3 DDR5 (Pc-1, Dc-1S)

| Parameter | Value | Notes |
|---|---|---|
| VDDQ | 1.1 V | DDR5 standard |
| DQ drive current | 34 Ω / 40 Ω | ODT-selectable |
| DQS differential impedance | 80 Ω ±10% | --- |
| CA single-ended impedance | 40 Ω ±10% | (this design uses single-ended CA) |
| Maximum trace length | 100 mm | Per channel, with matched DQ/DQS |
| DQ-to-DQS skew | $< 10$ ps | Per byte lane |

### .45.4 LPDDR5X (Ph-1)

Same standard as DDR5, with 1.05 V VDDQ, on-package memory so traces are short (<20 mm).

### .45.5 Electrical summary

| Interface | Diff Z | VDDQ | Max trace |
|---|---|---|---|
| PCIe Gen5 / CXL | 85 Ω | 1.1 V (AC-coupled) | 300 mm |
| DDR5 | 80 Ω | 1.1 V | 100 mm |
| LPDDR5X | 80 Ω | 1.05 V | 20 mm |

## .46 Validation and Test

### .46.1 Ballmap validation

Before tape-out, the ballmap CSV is validated against:

1. Signal catalog consistency: the CSV ball counts per signal group match §E.3.
2. Diff pair completeness: every DiffPair name in the CSV has exactly 2 balls (one P, one N).
3. Rail assignment: every ball has either a Rail name or a Direction of NC.
4. Coordinate uniqueness: no ball coordinate appears twice.
5. Contiguity: the ball grid has no unexpected gaps.

The validation script (ballmap_validate.py) produces a report.

### .46.2 Package assembly validation (post-silicon)

After the first silicon returns, the package is validated:

1. X-ray inspection: confirms all 400/1851/4677 solder joints are present and well-formed.
2. Cross-section: confirms solder joint shape and standoff height.
3. Electrical continuity: confirms every signal ball is electrically connected to the die pad it is assigned to.
4. Diff pair impedance measurement: TDR measurement on representative diff pairs, confirming 85 Ω (PCIe) or 80 Ω (DDR5).
5. PDN impedance measurement: vector network analyzer sweep of the VDD_CORE plane, confirming target impedance.

### .46.3 Board-level acceptance

For a board design to be accepted, it must:

1. Reproduce the ballmap assignment exactly (no ball re-purposed).
2. Meet the diff pair impedance and trace length specifications.
3. Meet the DQ-to-DQS skew specification for all DDR5/LPDDR5X channels.
4. Meet the VDD_CORE PDN impedance target (typically $< 1$ mΩ from DC to 10 MHz).

# Appendix F: Mechanical Drawings

This appendix specifies the mechanical dimensions, tolerance stack-up, heatsink mounting, thermal interface, and board keep-out zones for the three packages. The mechanical drawing package (DXF, DWG, SVG, and PDF files) ships with this specification; this appendix gives the dimensions in tabular form.

## .47 Package Outline Dimensions

### .47.1 FCBGA 16×16 (Ph-1)

| Dimension | Nominal | Tolerance | Notes |
|---|---|---|---|
| Package body length | 16.00 mm | ±0.10 mm | X-dimension |
| Package body width | 16.00 mm | ±0.10 mm | Y-dimension |
| Package body thickness | 1.00 mm | ±0.05 mm | Substrate + lid |
| Substrate thickness | 0.60 mm | ±0.03 mm | --- |
| Lid thickness | 0.40 mm | ±0.03 mm | Ni-plated Cu |
| Ball array field | 10.00 mm × 10.00 mm | ±0.05 mm | 20×20 grid |
| Ball pitch | 0.50 mm | ±0.02 mm | --- |

| Dimension | Nominal | Tolerance | Notes |
|---|---|---|---|
| Ball diameter (pre-reflow) | 0.30 mm | ±0.02 mm | --- |
| Ball height (post-reflow) | 0.25 mm | ±0.03 mm | --- |
| Coplanarity | --- | ≤0.05 mm | All balls |
| Package weight | 0.9 g | ±0.1 g | --- |

The package has a laser-marked orientation dot at A1 corner (0.5 mm diameter) on the lid top surface. The pin 1 indicator is a chamfered corner on the substrate (45° chamfer, 1.0 mm).

### .47.2 LGA 1700 (Pc-1)

| Dimension | Nominal | Tolerance | Notes |
|---|---|---|---|
| Package body length | 45.00 mm | ±0.10 mm | X-dimension |
| Package body width | 37.50 mm | ±0.10 mm | Y-dimension |
| Package body thickness (with lid) | 3.50 mm | ±0.10 mm | --- |
| Substrate thickness | 1.20 mm | ±0.05 mm | --- |
| Lid thickness | 1.30 mm | ±0.05 mm | Ni-plated Cu, IHS |
| IHS-to-die TIM thickness | 0.20 mm | ±0.05 mm | --- |
| Ball array field | 41.60 mm × 33.80 mm | ±0.10 mm | --- |
| Ball pitch | 1.00 mm | ±0.03 mm | --- |
| Ball diameter (pre-reflow) | 0.60 mm | ±0.03 mm | --- |
| Package weight (with IHS) | 28 g | ±2 g | --- |

The package has an integrated heat spreader (IHS) with a stamped dimple for IHS-to-die thermal contact. The IHS is Ni-plated Cu, 1.3 mm thick, with a flatness of 0.05 mm over the TIM contact area.

The orientation marker is a triangular notch on the substrate edge at A1 corner (2.0 mm), and a laser-marked logo on the IHS top surface.

### .47.3 LGA 4677 (Dc-1S)

| Dimension | Nominal | Tolerance | Notes |
|---|---|---|---|
| Package body length | 61.00 mm | ±0.15 mm | X-dimension |
| Package body width | 82.00 mm | ±0.15 mm | Y-dimension |

| Dimension | Nominal | Tolerance | Notes |
|---|---|---|---|
| Package body thickness (with lid) | 3.80 mm | ±0.10 mm | --- |
| Substrate thickness | 1.40 mm | ±0.05 mm | --- |
| Lid thickness | 1.20 mm | ±0.05 mm | Ni-plated Cu, IHS |
| IHS-to-die TIM thickness | 0.25 mm | ±0.05 mm | --- |
| Ball array field | 56.00 mm × 76.00 mm | ±0.10 mm | --- |
| Ball pitch | 1.00 mm | ±0.03 mm | --- |
| Ball diameter (pre-reflow) | 0.60 mm | ±0.03 mm | --- |
| Package weight (with IHS) | 65 g | ±5 g | --- |

The package uses the standard Intel LGA 4677 socket specification (Eagle Stream platform). The IHS has the standard Intel server IHS pattern with the heatsink mounting features described in §F.3.

## .48 Substrate Stack-Up

### .48.1 FCBGA 16×16 substrate

| Layer | Thickness | Notes |
|---|---|---|
| L1 (top) | 15 µm Cu | Die attach + lid seal |
| L2 | 30 µm dielectric + 15 µm Cu | --- |
| L3 | 30 µm dielectric + 15 µm Cu | --- |
| L4 (core) | 400 µm BT resin | Rigid core |
| L5 | 30 µm dielectric + 15 µm Cu | --- |
| L6 (bottom) | 15 µm Cu + ENIG | Ball pads |
| Total | 600 µm | --- |

### .48.2 LGA 1700 substrate

8-layer build-up substrate: 1-2-1 (one core, two build-up each side). BT resin core 800 µm, plus 100 µm each side build-up = 1000 µm total. LGA pad finish: ENIG (Au 0.05 µm over Ni 3--6 µm).

### .48.3 LGA 4677 substrate

10-layer build-up substrate: 2-4-2 (one thick core, four layers each side). Total thickness 1400 $\mu$m. Same LGA pad finish.

## .49 Heatsink Mounting

### .49.1 FCBGA 16×16 (Ph-1)

The package has no separate heatsink mounting. The phone OEM attaches a thin vapor chamber or heat spreader directly to the lid via the TIM interface. The mounting is by adhesive or snap-fit, not by screws.

- TIM contact area: 12.0 mm × 12.0 mm (the lid top surface, minus a 2 mm border for seal).
- TIM thickness: 0.10--0.20 mm.
- Mounting pressure: 5--15 psi (typical for adhesive-based phone cooling).

### .49.2 LGA 1700 (Pc-1)

The package uses the standard Intel LGA 1700 heatsink mounting hole pattern (ILM, Independent Loading Mechanism). The board should provide:

- 4 mounting holes on a rectangular pitch, 75.0 mm × 82.0 mm (±0.10 mm).
- Hole diameter: 3.5 mm (±0.05 mm).
- Board mounting pad: 6.0 mm diameter (for M3.5 screws).
- Mounting pressure: 30--50 lbf (typical for 360 mm AIO), applied via the AIO backplate.

### .49.3 LGA 4677 (Dc-1S)

The package uses the standard Intel LGA 4677 heatsink mounting hole pattern (Eagle Stream socket):

- 4 mounting holes on a rectangular pitch, 100.0 mm × 115.0 mm (±0.15 mm).
- Hole diameter: 4.0 mm (±0.05 mm).
- Board mounting pad: 7.0 mm diameter (for M4 screws).
- Mounting pressure: 50--80 lbf (typical for cold-plate cooling).
- Torque specification: 8--10 in-lb per screw (M4 stainless steel, 4 screws).

## .50 Thermal Interface Material (TIM)

### .50.1 TIM specification

The package lid is supplied with no factory-applied TIM. The user applies TIM between the lid and the heatsink/cold-plate.

| Property | Value | Tolerance | Notes |
|---|---|---|---|
| Thermal conductivity | $\geq$ 8 W/m·K | --- | For Pc-1 (high TDP) |
| Thermal conductivity (DC) | $\geq$ 12 W/m·K | --- | For Dc-1S |
| Thermal conductivity (Phone) | $\geq$ 3 W/m·K | --- | For Ph-1 |
| Thickness (applied) | 0.05--0.20 mm | --- | Based on TIM type |
| Operating temperature | $-40$ to $+150$ °C | --- | --- |
| Interface resistance | $\leq$ 0.05 K·cm$^2$/W | --- | For 8 W/m·K paste |

### .50.2 TIM application

For Pc-1 and Dc-1S, apply TIM paste as a 5-dot pattern (center + 4 corners) on the IHS top surface, then mount the cooler. The mounting pressure spreads the TIM to 0.05--0.15 mm uniform thickness. Do not apply more than 0.5 g of TIM; excess can bridge to the substrate edge.

For Ph-1, apply TIM pad (0.15 mm thickness) between the lid and the vapor chamber.

## .51 Solder Reflow Profile

### .51.1 Reflow profile for board assembly

Standard SAC305 lead-free reflow, with the following profile (measured at the ball):

| Phase | Temperature | Time |
|---|---|---|
| Preheat | 25 $\rightarrow$ 150 °C | 60--90 s (max 3 °C/s ramp) |
| Soak | 150 $\rightarrow$ 200 °C | 60--120 s |
| Reflow | Peak 245 $\pm$ 5 °C | 20--40 s above 217 °C (liquidus) |
| Cooling | 245 $\rightarrow$ 25 °C | 60--90 s (max 6 °C/s down-ramp) |
| Total cycle | --- | 200--330 s |

### .51.2 Reflow constraints

- Peak temperature: 245 °C max. Higher peak increases intermetallic growth and reduces joint reliability.
- Time above liquidus: 20--40 s. Longer causes excessive IMC formation.

- Ramp rates: $\leq$3 °C/s up, $\leq$6 °C/s down. Faster causes thermal shock and potential die cracking.
- Number of reflows: The chip can survive 3 reflows (for double-sided assembly + rework). Pre-wake-up applied at factory test is designed to withstand 3 reflows.
- MSL: MSL 3 (168 hours floor life at 30 °C / 60% RH after bake). Higher MSL requires dry-pack storage.

## .52 Moisture Sensitivity Level (MSL)

| Package | MSL | Floor life | Notes |
|---|---|---|---|
| FCBGA 16×16 | MSL 3 | 168 h | Standard |
| LGA 1700 | MSL 3 | 168 h | Standard |
| LGA 4677 | MSL 3 | 168 h | Standard |

The packages ship in dry-pack (moisture barrier bag + desiccant + HIC). The bag is opened only when the board is ready to reflow. After opening, the floor life of 168 hours at $\leq$30 °C / $\leq$60% RH applies. If exceeded, the package is baked at 125 °C for 24 hours before use.

## .53 Board Keep-Out Zones

The board layout must respect the following keep-out zones (no traces, no vias, no components within the zone):

### .53.1 FCBGA 16×16 (Ph-1)

| Zone | Dimensions | Notes |
|---|---|---|
| Top (component) side | 0.3 mm from package outline | Underfill fillet space |
| Bottom (board) side | 0.2 mm from ball array edge | --- |
| Corner keep-out | 2.0 mm × 2.0 mm at each corner | Alignment mark |
| Board edge | 1.0 mm from package to board edge | For de-paneling |

### .53.2 LGA 1700 (Pc-1)

| Zone | Dimensions | Notes |
|---|---|---|
| Top side (component) | 5.0 mm from package outline | ILM backing plate area |
| Bottom side (board) | 3.0 mm from ball array edge | --- |
| Backplate area | 95 mm × 95 mm centered on package | For AIO backplate |
| Standoff height | 6.5 mm ± 0.3 mm | From board top surface to IHS top |

### .53.3 LGA 4677 (Dc-1S)

| Zone | Dimensions | Notes |
|---|---|---|
| Top side (component) | 8.0 mm from package outline | Cold plate clearance |
| Bottom side (board) | 4.0 mm from ball array edge | --- |
| Cold plate backplate | 130 mm × 130 mm centered | For cold-plate mounting |
| Standoff height | 8.0 mm ± 0.5 mm | From board top to IHS top |

## .54 Mechanical Stress and Reliability

### .54.1 Thermal cycling

The package must survive −40 to +125 °C thermal cycling, 1000 cycles, no mechanical or electrical failure.

### .54.2 Drop and shock

For Ph-1 (phone product): must survive 1.5 m drop onto concrete, 26 orientations, no electrical failure. The phone OEM encloses the package in its chassis; the chassis handles the drop energy.

For Pc-1 and Dc-1S: no drop test (server and desktop form factors).

### .54.3 Vibration

All three packages must survive 5--500 Hz random vibration at 1.5 g RMS, 30 min per axis, no failure.

## .55 Assembly Drawings Package

The mechanical drawing package ships separately and contains:

- fhpf_ph1_package.dxf | full PCB footprint, top-side component, and keep-out
- fhpf_pc1_package.dxf | same
- fhpf_dc1s_package.dxf | same
- fhpf_ph1_package_stackup.pdf | substrate stack-up drawing
- fhpf_pc1_package_stackup.pdf | same
- fhpf_dc1s_package_stackup.pdf | same
- fhpf_heatsink_pc1.dwg | recommended heatsink reference design
- fhpf_coldplate_dc1s.dwg | recommended cold-plate reference design
- fhpf_dimensions_check.pdf | all dimensions with GD&T callouts

The DXF files are the board designer's authoritative reference for the physical footprint. The PDF stack-up drawings give the exact layer composition, prepreg types, and copper weight. The heatsink/cold-plate DWG files are provided as reference designs; the system integrator may substitute alternative cooling solutions that meet the mechanical and thermal specifications in this appendix.

## .56 Summary Table

| Parameter | Ph-1 | Pc-1 | Dc-1S |
|---|---|---|---|
| Body size | 16×16×1.0 mm | 45×37.5×3.5 mm | 61×82×3.8 mm |
| Ball pitch | 0.50 mm | 1.00 mm | 1.00 mm |
| Ball count | 400 | 1851 | 4677 |
| IHS | thin lid | IHS | IHS |
| Heatsink mounting | adhesive | LGA 1700 ILM | LGA 4677 (Eagle Stream) |
| Max TDP | 32 W | 323 W | 430 W |
| MSL | 3 | 3 | 3 |
| Solder reflow peak | 245 °C | 245 °C | 245 °C |
| Reflow cycles max | 3 | 3 | 3 |

| Parameter | Ph-1 | Pc-1 | Dc-1S |
|---|---|---|---|
| TIM thermal cond. | $\geq$3 W/mK | $\geq$8 W/mK | $\geq$12 W/mK |

# Appendix G: Bring-Up Checklist

This appendix is the acceptance test that runs when the first silicon returns from the foundry. It defines, in order, every measurement that confirms the silicon is functional, with acceptance criteria, failure diagnosis, and recovery action for each. It is the checklist a bring-up engineer follows on the bench.

The checklist is organized in 12 stages. Each stage has a precondition (the prior stage passed), a list of measurements, and an exit criterion. A stage that fails blocks all later stages; the failure diagnosis in each stage describes the most likely causes and the recovery actions.

Every measurement is stated with the instrument (oscilloscope, multimeter, logic analyzer, etc.), the pin or register under test, and the numeric acceptance criterion. Nothing in this appendix is optional: all 12 stages must pass before silicon is deemed functional.

## .57 Stage 1: Unpowered Continuity and Isolation

Preconditions: board assembled, chip not powered, no power supply connected.

Measurements:

1. VDD_CORE short to VSS: multimeter in resistance mode, measure from any VDD_CORE ball to any VSS ball. Acceptance: $> 10\ \Omega$.
2. VDD_IO short to VSS: same. Acceptance: $> 10\ \Omega$.
3. VDD_ANALOG short to VSS: same. Acceptance: $> 10\ \Omega$.
4. VDD_PLL short to VSS: same. Acceptance: $> 10\ \Omega$.
5. VDD_CORE short to VDD_IO: measure between rails. Acceptance: $> 1$ k$\Omega$.
6. VDD_CORE short to VDD_ANALOG: same. Acceptance: $> 1$ k$\Omega$.
7. Signal ball isolation: select 10 signal balls at random (including at least one diff pair per interface). Measure to VSS. Acceptance: $> 1$ M$\Omega$.
8. Socket contacts: for LGA 1700 and LGA 4677, visually inspect socket pins under 20$\times$ magnification; bent pins are rejects.

Exit criterion: all 8 measurements pass.

Failure diagnosis:

- Short on a rail: board assembly defect (solder bridge, misplaced component). BGA rework required.

- Short between rails: package defect or board short. Chip replacement likely required.
- Short on a signal: board assembly defect or chip ESD damage.

## .58 Stage 2: Rail Bring-Up (No Chip Load)

Preconditions: Stage 1 passed; chip on board but not powered.
Procedure: connect bench supply to VDD_CORE only, with current limit at 100 mA. No other rails powered. Set voltage to 0.50 V.
Measurements:

1. Current draw at 0.50 V: acceptance: $< 50$ mA.
2. VDD_CORE voltage: measured at the chip ball (Kelvin connection). Acceptance: 0.50 $\pm$ 0.02 V.
3. Repeat for each rail: VDD_IO (1.8 V, acceptance $< 100$ mA), VDD_ANALOG (1.5 V, acceptance $< 50$ mA), VDD_PLL (1.0 V, acceptance $< 20$ mA).

Exit criterion: current draws within specification at nominal rail voltages.
Failure diagnosis:

- Excessive current at 0.5 V: chip defect (latch-up, ESD damage) or board short. Chip replacement.
- Excessive current at nominal VDD_ANALOG: analog block defect. Chip replacement.
- Excessive current at VDD_PLL: PLL defect. Chip replacement.

## .59 Stage 3: Power Sequencing

Preconditions: Stage 2 passed. All rails connected; bench supply replaced with board VRMs.
Procedure: apply main power input; measure each rail's rise time and sequence order against the specification in Appendix A §A.11.
Measurements:

1. Rail 1 (VDD_IO) rise time: acceptance: 0.1--10 ms, monotonic.
2. Rail 2 (VDD_ANALOG) rise time: acceptance: 0.1--10 ms, monotonic, after Rail 1 (delay 1--10 ms).
3. Rail 3 (VDD_PLL) rise time: acceptance: 0.1--5 ms, after Rail 2.
4. Rail 4 (VDD_CORE) rise time: acceptance: 0.1--50 ms, after Rail 3.
5. POR release: RESET pin rises after all rails are within 5% of nominal. Acceptance: POR rise time 1--100 ms after last rail OK.
6. Rail_STATUS register: read via JTAG (no CPU yet). Acceptance: RAIL_STATUS = 0x3F (all 6 rails).

Exit criterion: all 6 measurements pass.
Failure diagnosis:

- Wrong sequencing: PMIC or board sequencing circuit defect. Rework.
- Rail rise time out of spec: VRM design issue. Rework.
- Rail_STATUS not all 6 bits: some rail not OK. Check that rail's VRM.

## .60 Stage 4: PLL Lock and Clock Distribution

Preconditions: Stage 3 passed. Chip is powered, POR released.
Procedure: check the compute PLL and the CXL LC-PLL lock via JTAG status register; measure clock at the package pin.
Measurements:

1. Compute PLL lock: read via JTAG (offset 0x4000_9000 register 0x20, bit 12). Acceptance: LOCK = 1 within 10 ms of POR release.
2. Compute PLL frequency: measure at a clock test point (if the package brings one out; otherwise measure via the CPU clock counter over a known wall-clock interval). Acceptance: 6.20 $\pm$ 0.03 GHz on Pc-1, 4.50 $\pm$ 0.03 GHz on Ph-1, 6.00 $\pm$ 0.03 GHz on Dc-1S.
3. CXL LC-PLL lock (Pc-1, Dc-1S): read via JTAG register 0x4000_9000 register 0x24, bit 12. Acceptance: LOCK = 1.
4. Clock jitter: measure via spectrum analyzer (if external clock is available) or via the CPU clock noise floor. Acceptance: $<$1.0 ps RMS integrated 100 Hz--100 MHz.

Exit criterion: all measurements pass.
Failure diagnosis:

- PLL not locking: PLL reference absent or wrong frequency. Check the reference crystal / oscillator on the board.
- PLL locking at wrong frequency: divider misconfiguration. Check the boot OTP settings or the strapping pins.
- Jitter $>$ 1 ps: PLL design margin; check the reference quality and VDD_PLL noise.

## .61 Stage 5: JTAG and Boot ROM Execution

Preconditions: Stage 4 passed. PLL locked.
Procedure: connect JTAG; halt the boot hart at the boot ROM entry point; step through the first 100 instructions; confirm the boot ROM signature and the first-branch target.
Measurements:

1. JTAG TAP IDCODE: acceptance: reads the expected device ID (0x1D1F_0001 for Ph-1, 0x1D1F_0002 for Pc-1, 0x1D1F_0003 for Dc-1S).
2. Boot ROM entry: after POR + JTAG halt, PC = 0x0000_0000_0000_0000. Acceptance: exact.
3. First instruction: read memory at 0x0. Acceptance: matches the boot ROM image.

4. Boot ROM signature: read the last 16 bytes of the boot ROM region. Acceptance: matches the expected ECDSA-public-key hash (readable at 0x0000_0000_0000_0800).

5. First branch target: step to the first ‘j‘ or ‘jal‘ instruction; confirm the target is inside the boot ROM address range (0x0 to 0x1000).

Exit criterion: all 5 measurements pass.
Failure diagnosis:

- JTAG IDCODE wrong: chip defect (mask ROM or TAP issue) or JTAG pin damage. Chip replacement.
- Boot ROM entry not 0: boot strapping or boot selection strapping pins wrong. Check board straps.
- Boot ROM signature mismatch: die-level mask issue. Chip replacement.

## .62 Stage 6: Boot ROM Completion

Preconditions: Stage 5 passed. JTAG verified boot ROM start.
Procedure: remove JTAG halt; allow the boot ROM to run to completion; capture the boot ROM UART output.
Measurements:

1. UART output at 115200 8N1: acceptance: firmware banner appears within 250 ms of POR release.
2. DDR5 training status: captured in the UART log. Acceptance: "DDR5 training: PASS."
3. PCIe enumeration (Pc-1, Dc-1S): captured in the UART log. Acceptance: "PCIe: N devices enumerated" with $N \geq 1$ (assuming a PCIe device is populated).
4. CXL enumeration (Pc-1, Dc-1S, multi-socket only): captured in the UART log. Acceptance: "CXL: M remote sockets" with $M \geq 1$ (single-socket: $M = 0$).
5. ACPI tables or DT blob loaded: captured in the UART log. Acceptance: "Platform description: [ACPI | DT] at address 0xXXXXXXXX."
6. Jump to OS loader: captured in the UART log. Acceptance: "Jump to OS loader."

Exit criterion: all 6 measurements pass.
Failure diagnosis:

- No UART output: UART pin misrouted; UART clock absent; boot ROM hung. Check Stage 5.
- DDR5 training fail: memory signal integrity; check the DDR5 trace length and impedance (Appendix E §E.10).
- PCIe enumeration 0: PCIe device not populated or link training fail. Check the PCIe diff pair polarity.
- ACPI/DT not loaded: boot OTP boot-mode bit wrong.

## .63 Stage 7: DDR5 Memory Test

Preconditions: Stage 6 passed. DDR5 training passed; boot ROM passed to OS loader.
Procedure: halt at the OS loader entry; run a memory test utility (memtest86+, or the design-supplied dramtest) over the full DRAM range.
Measurements:

1. Memtest86+ full pass: acceptance: 0 errors over the full memory.
2. Bandwidth: measure with stream or mbw. Acceptance: $\geq$400 GB/s on Pc-1 (8 channels), $\geq$200 GB/s on Dc-1S (4 channels), $\geq$100 GB/s on Ph-1 (2 LPDDR5X channels).
3. Latency: measure with lmbench or Intel MLC. Acceptance: 38--54 ns.
4. Refresh and ECC: enable ECC, inject single-bit errors via register writes; confirm correction. Acceptance: single-bit corrected, double-bit detected.

Exit criterion: all 4 measurements pass.
Failure diagnosis:

- Memtest errors: DRAM defective, DRAM timing marginal, DDR5 signal integrity. Check with reduced clock (DDR5-5600 instead of 7200) to isolate.
- Low bandwidth: one channel not training; check DDR5 channel enable in the boot ROM log.
- High latency: DDR5 training marginal; retrain at lower frequency.

## .64 Stage 8: NVL3 First-Program and Read-Back

Preconditions: Stage 7 passed. DDR5 functional. NVL3 controller initialized per Appendix B §B.4.
Procedure: run the NVL3 first-program test. This test writes a known pattern (walking-1, walking-0, checkerboard, all-0, all-1) to every bank, reads back, verifies, and checks ECC status.
Measurements:

1. NVL3 calibration: confirm CALIB_STATUS.DONE = 1 after the boot path's CALIB_START sequence. Acceptance: done within 20 ms.
2. NVL3 first-program test: acceptance: 0 errors in the pattern write/read-back on all banks.
3. NVL3 column leakage: read the column leak screening per Appendix A §A.16. Acceptance: $>$99% of columns classified Normal; reduced-margin columns reported.
4. NVL3 ECC single-bit: inject via register; confirm correction. Acceptance: corrected, address logged.
5. NVL3 ECC double-bit: inject via register; confirm detection. Acceptance: detected, address logged, error reported.
6. NVL3 retention: program known pattern, power down for 1 minute (post-factory pre-wake-up should be present), power back up, read back. Acceptance: 0 errors.

Exit criterion: all 6 measurements pass.
Failure diagnosis:

- Calibration not done: OTP calibration table corrupt. Rerun factory calibration.
- First-program errors: HZO film phase fraction low (see Appendix A §A.16). Check Gate 1 acceptance.
- High column leakage: process marginality. Column screening will flag bad columns.
- Retention fail without pre-wake-up applied: pre-wake-up missing from this die. Rerun factory test.

## .65 Stage 9: Crossbar First-Program and Calibration (Ph-1, Pc-1)

Preconditions: Stage 8 passed. NVL3 functional. Crossbar controller initialized per Appendix B §B.5.

Procedure: load the calibration table from OTP; program a known weight pattern; run a MAC operation; compare against the expected analog output.

Measurements:

1. Crossbar calibration: confirm CALIB_STATUS.DONE = 1. Acceptance: done within 5 ms.
2. DAC monotonicity: iterate all 16 DAC codes; measure ADC output; acceptance: 16 distinct outputs in increasing order.
3. Weight program: program a row with pattern 00, 01, 10, 11 repeated; read back; acceptance: all cells correct.
4. ADC ENOB: apply a known input ramp; compute ENOB. Acceptance: $\geq$8.5 bits (spec 8.9).
5. MAC operation: program a 16×16 identity matrix; apply a 16-element input vector; compare output to input. Acceptance: output = input within 2 LSB (for each output element).

Exit criterion: all 5 measurements pass.
Failure diagnosis:

- Calibration fail: OTP calibration table corrupt or TIA/ADC marginal. Rerun factory calibration.
- DAC monotonicity fail: DAC design defect. Chip replacement.
- Weight program errors: FeFET cell defect or write-verify controller issue. Rerun first-program.
- ENOB fail: ADC design or VDD_ANALOG noise. Check the analog rail.
- MAC mismatch: crossbar analog chain misconfigured. Re-run calibration.

## .66 Stage 10: PCIe Enumeration and DMA

Preconditions: Stage 9 passed (or Stage 8 for Dc-1S). Platform description loaded.
Procedure: after boot into OS loader, run PCIe enumeration. Plug in a known-good NVMe drive; run a PCIe bandwidth test.
Measurements:

1. PCIe enumeration: lspci lists all populated PCIe devices with correct vendor/device IDs. Acceptance: device count matches physical population.
2. PCIe link width: lspci -vv shows the expected width (x16 for the main PCIe slot; x4 for the NVMe slot). Acceptance: matches physical slot.
3. PCIe link speed: shows the expected generation (Gen5 on Pc-1 and Dc-1S; Gen3 on Ph-1). Acceptance: 32 GT/s (Gen5) or 8 GT/s (Gen3).
4. PCIe DMA bandwidth: dd read from the NVMe at sustained queue depth. Acceptance: $\geq$12 GB/s for Gen5 x4, $\geq$3 GB/s for Gen3 x2.

Exit criterion: all 4 measurements pass.
Failure diagnosis:

- Device not enumerated: board-level PCIe routing issue (diff pair polarity, length mismatch), or PHY configuration issue. Check diff pair impedance (Appendix E §E.10).
- Wrong width: one or more lanes not training. Check per-lane status in the CXL control register (Appendix A §A.12).
- Wrong speed: link training failing at Gen5; retrain at Gen4 or Gen3 to isolate.

## .67 Stage 11: Kernel Boot to Userspace

Preconditions: Stage 10 passed (Dc-1S, Pc-1) or Stage 9 passed (Ph-1). OS image prepared.
Procedure: boot into a full Linux kernel; measure boot time, check dmesg for errors, reach userspace.
Measurements:

1. Kernel boot time to userspace: acceptance: $\leq$500 ms (spec is 400 ms with 20% margin).
2. dmesg errors: acceptance: no "ERROR" or "Call trace" lines (warnings may be present).
3. CPU enumeration: lscpu shows the correct number of CPUs at the correct nominal frequency. Acceptance: matches product (10 on Ph-1, 28 on Pc-1, 32 on Dc-1S).
4. Memory visible: free -h shows the correct DRAM size. Acceptance: within 1% of physical population.
5. NVL3 visible: lsblk or /sys/block shows the NVL3 block device. Acceptance: capacity matches (64 MB on Ph-1/Pc-1, 24 MB on Dc-1S).
6. Thermal sensors: sensors reads all sensors with plausible temperatures (within $\pm$5 °C of ambient + offset). Acceptance: all sensors report values.

7. Crossbar visible (Ph-1, Pc-1): /dev/crossbar device present. Acceptance: device node exists.

Exit criterion: all 7 measurements pass.
Failure diagnosis:

- Boot time >500 ms: DRAM training slow, or ACPI enumeration slow. Check Stage 7 timing.
- dmesg errors: driver initialization failures. The UART log at the failing driver identifies the block.
- Wrong CPU count: MADT/DT issue. Check Appendix D.
- NVL3 invisible: NVL3 driver not loaded. Check the kernel config and the DSDT/DT NVL0 node.
- Sensors not readable: thermal driver issue. Check Appendix A §A.6 register map.

## .68 Stage 12: Userspace Verification and Application Run

Preconditions: Stage 11 passed. Userspace boot complete.
Procedure: run a representative workload to confirm end-to-end functionality.
Measurements:

1. CPU stress test: stress-ng --cpu N --timeout 60s. Acceptance: no errors, no thermal throttle above threshold.
2. Memory stress: stress-ng --vm 4 --vm-bytes 4G --timeout 60s. Acceptance: no OOM, no errors.
3. NVL3 read/write: dd if=/dev/nvl3 of=/dev/null bs=1M and reverse. Acceptance: read at line-speed, write without verify error.
4. Crossbar AI (Ph-1, Pc-1): run ResNet-50 inference on the crossbar test image. Acceptance: accuracy $\geq$75.5% (spec 75.7%, margin 0.2 pp).
5. Windows application (Pc-1, Dc-1S): run Firefox for Windows under Wine+Box64. Acceptance: launches, navigates to a web page, no crash.
6. Sleep/wake (P3$\rightarrow$P0): issue sleep, measure wake time. Acceptance: wake time $\leq$15 µs.
7. Cold boot from NVL3: power cycle, measure boot time. Acceptance: $\leq$500 ms.

Exit criterion: all 7 measurements pass.
Failure diagnosis:

- CPU stress failure: CPU microarchitecture bug. Further diagnosis with perf.
- Crossbar accuracy low: re-run calibration (Stage 9); if still low, crossbar cells defective.
- Windows app crash: Wine configuration or Box64 issue. Check the compatibility matrix.
- Sleep/wake slow: NVL3 state not retained, or PLL relock slow. Check the P3 wake path.

## .69 Bring-Up Summary

All 12 stages must pass before the silicon is deemed functional. Each stage blocks the next. The total bench time, once a board is assembled and a debug setup is in place, is approximately 4--6 hours for a first-pass run, or 1--2 days if any stage requires debugging. After the first successful bring-up, subsequent boards take 30--60 minutes per board.

The bring-up log (every measurement with the measured value, not just pass/fail) is a required deliverable. It becomes the reference dataset for future silicon revisions.

## .70 Bring-Up Equipment List

| Equipment | Purpose |
|---|---|
| Bench power supply (0--5 V, 20 A) | Stage 2 |
| Digital multimeter (4.5 digit) | Stages 1, 2, 3 |
| Oscilloscope (500 MHz, 4-channel) | Stages 3, 4 |
| Spectrum analyzer (26.5 GHz) | Stage 4 (jitter) |
| Logic analyzer (16-channel, 500 MS/s) | Stages 5, 6 |
| JTAG debugger (RISC-V compatible) | Stages 4, 5, 6, 8, 9 |
| UART-to-USB adapter (115200 baud) | Stages 6, 11 |
| Thermal camera | Stage 12 |
| PCIe NVMe drive (known-good) | Stage 10 |
| Bootable OS image (with NVL3 driver) | Stage 11 |
| Crossbar test image (ResNet-50) | Stage 12 |

# Appendix H: Security Provisioning Procedure

This appendix specifies how the security credentials (device-unique ECDSA key, OEM root key hash, debug public key, boot nonce, rollback counter) are generated, injected into OTP, locked, and tracked through the device lifecycle. It is the production procedure for the security provisioning step in factory test, plus the operational procedures for key lifecycle management after the device ships.

The OTP injection sequence itself is in Appendix B §B.7; this appendix provides the surrounding context: where the keys come from, how they are delivered, how the factory systems handle them, and how they are used after provisioning.

## .71 Security Lifecycle States

Every device transitions through four states, tracked by a provisioning status word in OTP region 5:

| State | Name | Condition | What is allowed |
|---|---|---|---|
| S0 | Unprovisioned | OTP root key region blank | Factory provisioning only; boot fails secure boot check |
| S1 | Provisioned | Root key, device key, debug key, nonce all programmed; OTP regions 0--4 locked | Normal operation; secure boot enforced |
| S2 | Commissioned | First end-user boot; attestation certificate issued | Normal operation; debug unlock via challenge-response |
| S3 | Decommissioned | Rollback counter set to max; OTP region 0 locked permanently | Boot only with decommission key; no secure boot |

The provisioning status word is at OTP region 5, word 0. It is read at boot; the boot ROM enforces the state transitions. A device that boots with state S0 (unprovisioned) jumps to a recovery image (physical presence required) rather than the normal firmware.

## .72 Key Generation

### .72.1 OEM root key

The OEM root key is an ECDSA P-256 key pair. It is generated by the OEM's security team on an HSM, in the following procedure:

1. HSM initialization: per the OEM's HSM vendor procedure (e.g., Thales Luna, Entrust nShield, AWS CloudHSM). At least 3 HSM administrators must be present (M-of-N quorum, typical M=2, N=3).
2. Root key generation: HSM: generate-ecdsa-p256 --label "FHP-ROOT-2026" --usage sign,verify --exportability none. The key is generated inside the HSM and never leaves it in plaintext.
3. Root key backup: 5 shares of the root key are stored in geographically separate vaults, encrypted under a per-share key. Recovery requires 3-of-5 shares (Shamir).

4. Root public key extraction: HSM: export-public-key --label "FHP-ROOT-2026" --out fhp_root_pub.der. This is the value injected into OTP region 0 at factory test (its hash is what the boot ROM anchors).

5. Root key sign: the OEM uses the root key to sign the OEM bootloader, kernel, initrd, and userspace images. The signing procedure uses the HSM to produce the ECDSA signature; the private key never leaves the HSM.

### .72.2 Device-unique key

The device-unique ECDSA P-256 key is generated by the factory test system at the time of provisioning:

1. Factory HSM generates a fresh ECDSA P-256 key for the device: HSM: generate-ecdsa-p256 --label "FHP-DEV-<serial>" --usage sign,verify --exportability wrapped.

2. The private key is exported wrapped under the device's OTP public key (see §H.3), so the wrapper can be opened only by the specific device's security engine.

3. The public key is stored in the factory database (Production Key Management System, PKMS) indexed by device serial.

4. The wrapped private key is delivered to the device via the factory programming interface.

5. The factory HSM zeroizes the plaintext private key after wrapping (the wrapped form remains in PKMS).

The device-unique private key is used for debug unlock challenge-response and attestation signature. It is never exported in plaintext from the device after provisioning.

### .72.3 Debug public key

The debug public key is the OEM's chosen ECDSA P-256 public key for the debugger. It is generated by the OEM and provided to the factory test system as a DER file. The factory injects it into OTP region 2. This key authorizes debug unlock; the OEM can rotate it by generating a new key and issuing a firmware update that authorizes the new key (the old key stays in OTP but firmware rejects unlock requests signed by it).

### .72.4 Boot nonce

The boot nonce is a 16-byte random value generated by the device's TRNG at provisioning time. It is used to derive the per-boot encryption keys (Appendix A §A.12). The TRNG must pass its self-tests (Appendix A §A.13 register TRNG_STATUS.HEALTH) before the nonce is used. If the TRNG health check fails, the device is flagged for factory RMA.

## .73 OTP Injection Sequence

The detailed OTP write sequence is in Appendix B §B.7. The surrounding factory procedure is:

1. The device is on the factory ATE at the security provisioning station.

2. The factory system confirms the device is in state S0 (OTP region 0 blank).

3. The factory system is connected via secure channel to the factory HSM and PKMS.

4. The factory system requests a device-unique key pair; the HSM generates it and returns the wrapped private key + public key.

5. The factory system reads the OEM root key hash (from the OEM-provided DER file) and the debug public key (from OEM).

6. The factory system reads the device's TRNG output (16 bytes) via the JTAG/SPI programming interface.

7. The factory system calls the OTP injection sequence (Appendix B §B.7) to write root key hash, device key (wrapped), debug public key, rollback counter = 1, boot nonce.

8. The factory system calls the OTP read-verify for every word written.

9. If all read-verifies pass, the factory system writes OTP region 5 word 0 to the S1 transition value (0x5052_4F56 = "PROV" in ASCII).

10. The factory system locks OTP regions 0--4 (irreversible).

11. The factory system logs the successful provisioning to PKMS: device serial, public key, timestamp, factory ID, operator ID.

12. The factory system moves the device to the next station.

If any step fails, the device is scrapped and the provisioning log records the failure reason. There is no retry on OTP write failures (OTP is one-time programmable).

## .74 Rollback Counter Initialization

The rollback counter starts at 1 at provisioning time. Each firmware update increments it. The counter is a 32-bit monotonic value stored in OTP region 3. Firmware rejects any update with a version number $\leq$ the current counter. The counter is never reset except by decommissioning.

The OTP storage for the counter is 8 bytes (64 bits), with the high 32 bits reserved. The counter increments in place: OTP bits are one-way programmable, so a counter increment writes additional bits (the counter effectively becomes a "one-time programmable multi-bit field" that stores the largest value written, since the read returns the bitwise-OR of all written values in an increment-only encoding). The encoding is binary: counter=1 writes bit 0; counter=2 writes bit 1; counter=3 writes bits 0 and 1 (already 1); etc. Increments work by writing the new value; the read returns the OR of all writes, which equals the highest value written (because the OR of a binary sequence 1, 2, 3, 4... has all bits set that any value set, and the highest value always has a superset of bits of any lower value). The OTP controller must implement this OR-on-write semantics for the rollback counter region specifically (the rest of OTP is write-once).

## .75 Debug Credential Delivery

After provisioning, the device is in state S1 (provisioned). The debug public key is in OTP, but the debugger also needs:

1. Device serial number: printed on the package top (2D barcode + human-readable). The factory PKMS holds the mapping serial $\rightarrow$ device public key.
2. Device public key: exported from the package via the SPI/I2C debug interface at provisioning time; also available from PKMS by serial.
3. Provisioning certificate: the OEM's root key signs the device public key, producing an X.509-like certificate that the debugger verifies.

The debug certificate is written to an EEPROM on the board (separate from the chip, on the factory-programmed EEPROM) at board assembly time, or sent to the OEM by PKMS (whichever flow the OEM prefers). The debugger reads the certificate from the EEPROM (or from the OEM's PKMS) and uses the certified device public key to verify the challenge-response.

## .76 Attestation Key Derivation

The device-unique ECDSA key is the root of the attestation chain, but it is not used directly for every attestation. Instead, the boot ROM derives a per-boot attestation key from the device key + boot nonce:

1. Read device private key from OTP (wrapped).
2. Unwrap in the security engine (never exits the security engine).
3. Read boot nonce from OTP.
4. Derive: attestation_key = HKDF-SHA-256(device_key, boot_nonce || 0x01, "FHP-AK", 32).
5. The attestation key is the ECDSA private key used to sign the DICE attestation quote.
6. The attestation public key is exported via the DICE CDI register (Appendix A §A.13); the OEM's root key signs the attestation public key into an attestation certificate.

The attestation certificate has a validity of 24 hours after boot. A verifier (e.g., a cloud service) checks: (a) the attestation certificate is signed by the OEM root key; (b) the attestation certificate's public key matches the signed quote; (c) the quote's boot measurement matches the expected boot image hash.

## .77 Debug Unlock Procedure (Operational)

The debug unlock procedure at any time after provisioning:

1. The debugger reads the device serial (visible on the package, or via SPI at bring-up).

2. The debugger retrieves the device public key from the OEM PKMS or from the board EEPROM.

3. The debugger generates a random 32-byte challenge.

4. The debugger writes the challenge to the OTP controller's CHALLENGE register (Appendix A §A.14).

5. The device security engine signs the challenge with the device private key and writes the response to RESPONSE register.

6. The debugger reads the response and verifies it against the device public key.

7. If verification succeeds, the debugger writes VERIFY=1 then UNLOCK=1 to CTRL.

8. The device opens the JTAG port for the configured TIMEOUT (default 1 hour).

9. All subsequent JTAG operations are logged to the device's event log (OTP region 6, 4 KB ring buffer).

Every failed verification attempt increments ATTEMPTS. After 5 failed attempts, LOCK_DOWN is set; the device permanently refuses JTAG unlock until a factory RMA reverses the counter (the RMA process involves an OEM-signed recovery image that resets the attempts counter once).

## .78 Decommissioning

When the device reaches end of life, or when the OEM determines the device must be permanently disabled (e.g., compromised credentials), the decommission sequence runs:

1. The OEM issues a "decommission" signed image that the device accepts only when it is signed by the OEM root key and has the special decommission flag.

2. The device, on running the decommission image, writes the OTP rollback counter to max (0xFFFF_FFFF) and sets OTP region 5 word 0 to the S3 transition value (0x4445_434F = "DECO").

3. OTP region 0 (root key hash) is left readable but no valid firmware image exists for it (the OEM root key is retired).

4. The device boots only into a stub recovery image that prints "DECOMMISSIONED" to UART and halts.

5. The OEM's PKMS marks the device as decommissioned; the device public key is revoked in the OEM's CRL.

The decommission is irreversible (OTP bits cannot be cleared). The device is functionally dead after decommission; it must be physically destroyed for end-of-life disposal per the OEM's e-waste procedure.

## .79 Security Event Logging

The device maintains an event log in OTP region 6 (4 KB ring buffer). Each entry is 32 bytes: timestamp (8 bytes), event code (1 byte), event-specific payload (23 bytes). Event codes:

| Code | Event | Payload |
|---|---|---|
| 0x01 | Boot successful | Boot image hash (truncated 8 bytes) |
| 0x02 | Boot signature failure | Expected hash, actual hash (truncated) |
| 0x03 | Rollback attempt blocked | Attempted version, current version |
| 0x04 | Debug unlock requested | Challenge (truncated 8 bytes) |
| 0x05 | Debug unlock succeeded | Session duration (seconds) |
| 0x06 | Debug unlock failed | Attempt number |
| 0x07 | Debug lock-down | --- |
| 0x08 | Tamper detected | Sensor (voltage/temp/clock) |
| 0x09 | Secure boot mode change | Old mode, new mode |
| 0x0A | Firmware update | Old version, new version |
| 0x0B | Decommission started | --- |
| 0x0C | Factory provisioning | Factory ID, operator ID |

The event log is write-once (OTP) and is read out at RMA for incident investigation. It cannot be cleared or modified. A device whose event log contains unexpected events (e.g., repeated failed unlocks) is flagged for RMA review even if functionally working.

## .80 Summary Table

| Item | OTP Region | Written When | Locked |
|---|---|---|---|
| OEM root key hash | 0 | Factory provisioning | Yes (at provisioning) |
| Device public key hash | 1 | Factory provisioning | Yes (at provisioning) |
| Device private key (wrapped) | 1 | Factory provisioning | Yes (at provisioning) |
| Debug public key | 2 | Factory provisioning | Yes (at provisioning) |

| Item | OTP Region | Written When | Locked |
|---|---|---|---|
| Rollback counter | 3 | On firmware update | Never (write-once) |
| Boot nonce | 4 | Factory provisioning | Yes (at provisioning) |
| Provisioning status | 5 | Factory provisioning | Yes (at provisioning) |
| Event log | 6 | On security events | Never (write-once ring) |
| Calibration tables | 7 | Factory test | Yes (at provisioning) |
| Reserved | 8+ | --- | --- |

## .81 Production Sign-Off

A device ships from the factory only when the following are all true:

1. All 12 bring-up stages (Appendix G) passed.
2. OTP provisioning status is S1 (provisioned).
3. All OTP regions 0--5 are locked.
4. The event log contains only the provisioning event (no errors).
5. The device's public key is in the OEM PKMS.
6. The provisioning certificate has been issued (or the OEM has confirmed the public key is in the database).
7. The device passed the final electrical test.

A device that fails any of these is either reworked (if rework is possible) or scrapped.

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